



Trinity College Dublin

Coláiste na Tríonóide, Baile Átha Cliath

The University of Dublin

School of Mathematics

Geometric Interpretations of Scattering Amplitudes

Edward Heeney

Supervised by Professor Ruth Britto

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Abstract

This work displays the new geometric ways of considering scattering amplitudes, namely the kinematic associahedron[1], and the fatgraph method[2]. Initially a brief overview will show how amplitudes are currently calculated, in $\text{Tr}(\phi)^3$ theory, in chapter 1. A mathematical breakdown of the kinematic associahedron, [3], will be presented in chapter 2, we build up from scratch the geometry required for our use. In chapter 3, we begin constructing the actual kinematic associahedron, showing that its forms contain the expected amplitudes, which through poles and factorisation, manifest locality and unitarity, respectively. In chapter 4, we show the fatgraph method, and how one “fattened” Feynman diagram can encode the amplitudes of all necessary planar Feynman diagrams, for the expected amplitude. Finally in chapter 5, we show the duality between the kinematic associahedron, and the fatgraph method, showing their explicit connection. Original work is done in bringing the three main bodies of work this capstone is based on, [3], [1], [2], into one concise, (almost) all encompassing work. Original examples of the methods are also seen throughout the paper.

⁰I have read and understood the plagiarism provisions in the General Regulations of the University Calendar for the current year, found at <http://www.tcd.ie/calendar>. I have also read and understood the guide, and completed the ‘Ready Steady Write’ Tutorial on avoiding plagiarism, located at <https://libguides.tcd.ie/academic-integrity/ready-steady-write>.
Edward Heeney, 7th of April

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“... There’s a different way of thinking about this very basics physics, where the foundation principles of space-time and quantum mechanics aren’t the stars of the show...

We’d like to find a different question to which amplitudes are the answer.”

-Nima Arkani-Hamed

Contents

1	Introduction	8
1.1	The current method	8
1.1.1	$n = 4$ Particle Scattering	9
1.1.2	Five-Leg Feynman Tree Diagram	10
1.2	How Does Locality and Unitarity Manifest themselves in the Current Method?	10
2	The Mathematical Background of the Scattering Geometries	12
2.1	Positive Geometries and their Canonical Forms Introduction	12
2.2	Projective Algebras and Algebraic Sets	13
2.3	Definition - D-dimensional Positive Geometry	13
2.3.1	Recursive relations of Positive Geometries	14
2.3.2	Pseudo-Positive Geometries	14
2.3.3	One-dimensional Positive Geometries	15
2.4	Triangulations of Positive Geometries	16
2.4.1	Triangulations of Pseudo-Positive Geometries	16
2.4.2	Signed Triangulations	16
2.5	Morphisms of Positive Geometries	17
2.6	Generalisation of Simplices	18
2.6.1	The Standard Simplex	18
2.6.2	A projective Simplex	19
2.7	Generalisation of Polytopes	20
2.7.1	Aside on Projective Polytopes	20
2.7.2	Continuation of projective Polytopes, With Euclidean Polytopes	21
2.7.3	Dual Polytopes	21
2.8	Simple polytopes	23
3	The Kinematic Associahedron	24
3.1	Planar Scattering Forms on Kinematic Space	24
3.1.1	Kinematic Space	24
3.1.2	Planar Kinematic Variables	25
3.1.3	The Planar Scattering Form	25
3.2	Kinematic Associahedron	27
3.2.1	The Associahedron from Planar Cubic Diagrams	27
3.2.2	The Kinematic Associahedron	28
3.3	Connection to Amplitudes in $\text{Tr}(\phi^3)$ Theory.	29
3.3.1	Examples	31

3.4	Factorisation	33
4	Fatgraph Method	36
4.1	New Formulation	36
4.2	Example $n = 4$	37
4.2.1	$n = 4$ Fatgraph	37
4.2.2	C_{13} Curve	37
4.2.3	$\mathbf{g}$ -vectors	38
4.2.4	Sub-mountainscape Diagrams	38
4.2.5	The Headlight Function	41
4.2.6	Aside on Tropicalisation	42
4.2.7	The C_{24} curve	42
4.3	Example $n = 5$	43
5	Connection Between Associahedron and Fatgraph	48
5.0.1	The Fatgraph Feynman-Fan, as the Kinematic Associahedron Dual	48
5.1	The Fatgraph Restores All Feynman Diagram Triangulation of the Associahedron	49
6	Conclusions	52
7	Appendix	53
7.1	Mathematic Background	53
7.1.1	Complete Mathematical Definition of Algebraic Varieties and Algebraic sets	53
7.1.2	Definition - Residue Operator	54
7.1.3	Forms on Projective Spaces	54
7.1.4	Note on Positive Geometries	55
7.2	Kinematic Associahedron	55
7.2.1	Vertex Coordinates For the Kinematic Associahedron	55
7.2.2	$n = 6$ Kinematic associahedron	56
7.2.3	U -Variables	56
7.3	Fatgraph Method	56
7.3.1	Another Way to View the Headlight Function	56

Background

This work attempts to be a complete, self-contained guide, up to tree-level Feynman diagrams, of calculating scattering amplitudes, of particle collisions using geometric interpretations, neglecting both quantum mechanics and space-time. This work initially was meant to be focused on curve integrals of scattering amplitudes, focusing on the paper [2]. This will be called the *fatgraph method* in this project. However, it was found while researching this topic that an incredibly rich mathematical structure, containing many, many years of research, lay beneath this neatly formulated approach. As such backtracking became necessary to understand the underlying complexity that props up the fatgraph formulation of computing scattering amplitudes. This led to learning about the amplituhedron [4], which is not discussed in this thesis, as it is not necessary for this specific interpretation, but is the first exploration into this kind of thinking. Next this research found exactly what the fatgraph formalism stands upon, the *kinematic associahedron*. This research found that there is a direct connection between the fatgraph method and the associahedron method. The associahedron was found to contain all the mathematical building blocks required to allow the fatgraph method to exist. However, it was not without heavy mathematical, in particular geometrical, rigour. This journey led to the structure this paper is written in now.

Chapter 1

Introduction

Feynman diagrams are and have been the predominant method through which physicists have computed scattering amplitudes for particle collisions since the late 1940s, which was introduced to greatly simplify the many path integrals previously needed to be calculated. This method involves summing over all the possible paths taken by the particles, through summing over all the possible Feynman diagrams. In doing so additional structures such as the bulk space time, the space containing the particles trajectories, as well as the Hilbert space, where the bulk spacetime evolution of the wave function occurs. These two structures formulate locality, from spacetime, and unitarity, from quantum mechanics. Computing scattering amplitudes in this manner is very computationally intense. The method that will be outlined in this project was conjured up from the observation that formulas, such as the Parke-Taylor formula and the BCFW recursion relations, show massive cancellations taking place when calculating many particle scattering processes. This prompted the authors of the main papers this thesis is based on, [2], [3], [1], to think that from the simplifications, the intrinsic nature of quantum mechanics and spacetime were complications to the process of calculating amplitudes, and there might be other ways to about this, while still to achieving unitarity and locality.

The basis of this work is to look into this new method of calculating scattering amplitudes, without reference to bulk-spacetime or the quantum mechanics.

1.1 The current method

Quickly we will note, that this project only focuses on tree-level diagrams, anything of a higher order is disregarded. Also we will be working $\text{Tr}(\phi^3)$ theory. We will be considering the massless and massive case of the this theory, for the kinematic associahedron and fatgraph method, respectively. We will be working in the $1/N$ limit. The $1/N$ limit means instead of working with $SU(3)$ and 3 colours, we are instead working with $SU(N)$ and N colours in the large N limit, which most importantly for us will allow us to only consider planar diagrams [5]. A specific colour ordering will be applied, to allow for cyclic permutations of external particle momenta. We will note that the theory we are working in is a scalar theory, meaning we do not have to consider spin or helicity or anything else in this regard. We will also be considering every particle as incoming.

1.1.1 $n = 4$ Particle Scattering

. As we are in $\text{Tr}(\phi^3)$ theory, there will only be cubic interactions at vertices.

In the current method of computing the amplitude for a four leg tree-diagram, in the non planar case, we construct our four-leg tree diagram, which can be seen in figure 1.1.

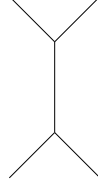


Figure 1.1: Four-leg Feynman tree diagram for $\text{Tr}(\phi^3)$ theory.

We can then label each external edge from 1 to 4. This single diagram will give us one possibility for our kinematic variable variables,

$$X_{i,j} = (p_i + \dots + p_{j-1})^2, \quad (1.1)$$

equivalently $X_{i,j} = (p_i + \dots + p_{j-1})^2 + m^2$ for the massive case. The k terms are the momenta associated with each external edge. However this is not the only possibility. From a cyclic permutation of labelling we end up with another four-leg Feynman tree diagram, which corresponds to another kinematic variable.

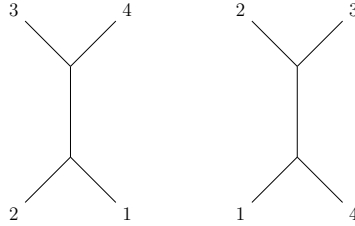


Figure 1.2: Four-leg Feynman tree diagram labels for $\text{Tr}(\phi^3)$ theory.

These are the only two possible four-leg Feynman tree diagrams in our theory, and any other possible cyclic permutation will give us a repeat of an already acquired kinematic variable. Thus defines the kinematic variables for both of the diagrams as

$$X_{1,3} = (p_1 + p_2)^2 \quad (1.2)$$

$$X_{2,4} = (p_2 + p_3)^2 \quad (1.3)$$

In this case we get the result for the amplitude as

$$\mathcal{A} = \frac{1}{X_{13}} + \frac{1}{X_{24}}. \quad (1.4)$$

Throughout this project, for the four particle case, we will label these kinematic variables in terms of the ordinary Mandelstam variables, s , t , and u . We will say that $X_{1,3} = (p_1 + p_2)^2 = s$

and $t = X_{2,4} = (p_2 + p_3)^2$. These kinematic variables, as you might have noticed, are not our typical s and t channels, where $s = (p_1 + p_2)^2$ and $t = (p_1 + 2_3)^2$. In our case we will make use of the fact that in the massless case $s + t + u = 0$, so our Mandelstam variables are able to be defined in terms of each other, and along with the fact we are considering all particles incoming, as well as all being scalar, the typical definitions of our s , t , and u channels become somewhat arbitrary.

1.1.2 Five-Leg Feynman Tree Diagram

The five-leg tree diagram is calculated in the same way as the four-leg tree diagram. We start with our initial five-leg Feynman diagram as seen in figure 1.3.



Figure 1.3: Five-leg Feynman diagram.

We then label our diagram initially as seen in figure 1.4, (a), and then cyclically permute our labelling as seen in (b)-(e). Our Kinematic terms follow the same definition as in $X_{i,j} = (p_i + \dots + p_{j1})^2 + m^2$, and using conservation of momentum at vertices, our amplitude is,

$$\mathcal{A}(5) = \frac{1}{X_{13}X_{14}} + \frac{1}{X_{14}X_{24}} + \frac{1}{X_{24}X_{25}} + \frac{1}{X_{25}X_{35}} + \frac{1}{X_{35}X_{13}} \quad (1.5)$$

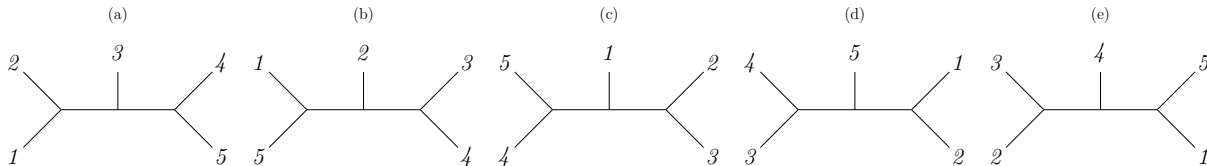


Figure 1.4: Five-Leg planar Feynman diagrams labelled ((a)-(e)).

1.2 How Does Locality and Unitarity Manifest themselves in the Current Method?

By the end of this project, we aim to show that the new geometric interpretations, can give scattering amplitudes, which also have manifest within them, the locality and unitarity of the quantum field theory approach, i.e. of Feynman diagrams. It is important to briefly demonstrate how locality and unitarity can be seen within the Feynman diagram method.

We will start by showing locality, in a non rigorous manner. There exists a concept in quantum field theory, which is that a particle can be “on-shell”. What this means, in natural units, is that the momentum squared of the particle, is equal to the mass squared, or equivalently zero, in the massless case. This is always the case of real, observable particle. Where locality enters the story is when virtual particles, i.e. particles with squared momenta of our Mandelstam variables, or more generalised kinematic variables, are considered. As we can

see in our $n = 4$ and $n = 5$ particle scattering case, when “off-shell” particles, (particles with the kinematic variables equal to the square of their momenta), are brought on shell, the amplitude now has a pole. In a physical sense we can essentially consider these poles to occur when the two particles, whose individual momenta are featured within our kinematic variables, meet at a point in space-time. We will consider this to be the locality manifesting within Feynman diagrams.

Now we will move on to show unitarity of Feynman diagrams. Briefly, unitarity of Feynman diagrams can be seen through factorisation. It is through this factorisation that we will claim that our kinematic Associahedron manifests unitarity. In quantum field theory, when we use factorising procedures, such as the optical theorem, we are working with the S matrix, which itself is a unitary operator. The S matrix itself is defined as $S = \mathbb{1} + iT$, where T is a complex matrix operator. We can imagine a system of many particles colliding, where all of those interactions are defined by our S matrix. Next we will manipulate the following equation to show how the S matrix displays amplitude factorisation.

$$\begin{aligned} S^\dagger S &= (\mathbb{1} - iT^\dagger)(\mathbb{1} + iT) \\ \implies \mathbb{1} &= \mathbb{1} - i(T^\dagger - T) + T^\dagger T \\ \implies i(T^\dagger - T) &= T^\dagger T. \end{aligned} \tag{1.6}$$

We can now think of the interaction of all our particles as having been factorised into two processes, one which occurs in accordance to T and one which occurs in accordance to $T^\dagger$. A much more rigorous view of this section can be found in [6].

Chapter 2

The Mathematical Background of the Scattering Geometries

The first step needed to understand in detail how we can encode amplitudes into geometric objects, is to first define these necessary objects, mainly positive geometries and simple polytopes. It is required for this chapter to include all required definitions, without too much extra, to understand what sort of geometric objects we are working with. This chapter is based heavily off the 2017 paper, “Positive geometries and Canonical forms” [3]. We will see how the formulas for the *canonical forms*, which encode our amplitudes, arise. We will also see how positivity plays a key role in this whole discussion. This section is for mathematical completeness of the associahedron method. It is not necessary to understand everything presented within this section to see how the kinematic associahedron method can encode amplitudes.

2.1 Positive Geometries and their Canonical Forms Introduction

Positivity is essential to the picture of our geometric interpretations. Positivity in our context is used moreso to describe non-negativity. This is as volumes of shapes are positive, and the amplitudes are encoded within the volume.

Intuitively we can think of positive geometries as shapes which have zero dimensional points on their boundaries, like at the corners of squares. For some uses we will also need to define a pseudo-positive geometry which we might roughly think of as a shape which may not have zero dimensional boundary points, like a circle. The important note for this distinction is the the zero-dimensional boundaries of the positive geometries will give the form defined on it poles, which is needed to show locality of the amplitude. The pseudo-positive geometry will not have these bounds.

The following definitions are in complex projective space, which will be important to note, as our differential form, which will encode our positive geometries, will be projective. This means it is projectively invariant, which is the geometric correspondence to amplitudes being Lorentz invariance in space-time.

2.2 Projective Algebras and Algebraic Sets

To start with, we are working in complex projective. We will let $\mathbb{P}^N$ denote our N -dimensional complex projective space, and have the projection map $\mathbb{C}^{N+1} \setminus \{0\} \rightarrow \mathbb{P}^N$. We will let $\mathbb{P}^N(\mathbb{R})$ then denote the image of the real $N + 1$ dimensional space, $\mathbb{R}^{N+1} \setminus \{0\}$ in $\mathbb{P}^N$.

We will let X be a *complex projective algebra*, meaning it is the set of solutions in $\mathbb{P}^N$, of a finite set of homogeneous polynomial equations. We will assume these polynomials are endowed with real coefficients. We will denote $X(\mathbb{R})$ as the real part of X , which is just the solution set of $\mathbb{P}(\mathbb{R})$, of the same set of equations.

A *semialgebraic set* in $\mathbb{P}(\mathbb{R})$ is a finite union of subsets, for which every subset is cut out by finitely many homogeneous, real polynomial equations, $\{x \in \mathbb{P}^N(\mathbb{R}) \mid p(x) = 0\}$, and also by homogeneous, real polynomial inequalities $\{x \in \mathbb{P}^N(\mathbb{R}) \mid q(x) > 0\}$.

There is a completely rigorous mathematical definition of everything described in this subsection in the appendix section 7.1.

2.3 Definition - D-dimensional Positive Geometry

All of the shapes which we will construct later on, will be positive geometries, however in kinematic space.

A *D-dimensional positive geometry* is defined to be a pair $(X, X_{\geq 0})$, where X is an irreducible complex projective variety of complex dimension D , and $X_{\geq 0} \subset X(\mathbb{R})$ is a non-empty orientated closed semi-algebraic set of real dimension D satisfying the following assumptions and recursive axioms.

Assumptions of Positive Geometries

Let $X_{>0}$ be the interior of the subspace $X_{\geq 0} \subset X(\mathbb{R})$ of dimension D , where $X_{\geq 0}$ is the closure of $X_{>0}$. $X(\mathbb{R})$ must be a D -dimensional algebraic variety. If $X_{>0}$ has more than one connected components, we thus can have multiple choices of orientation.

Let $\partial X_{\geq 0} := X_{\geq 0} \setminus X_{>0}$. Let $\partial X = \overline{\partial X_{\geq 0}}$ denote *Zariski closure* in X , which is defined as,

$$\partial X = \{x \in X \mid p(x) = 0 \text{ if } p(y) = 0 \text{ for all } y \in \partial X_{\geq 0}\}. \quad (2.1)$$

For homogeneous polynomial p . This means that for a polynomial vanishing on $\partial X_{\geq 0}$, the polynomial also vanishes on ∂X , where ∂ is the largest set for where this occurs. ∂X is a closed algebraic set of X . $C_i, C_2, \dots, C_r$ are let to be the co-dimension one irreducible set of ∂X , which means the subset is of dimension $D - 1$. We will define $C_{i,\geq 0}$ as the closure of the the interior of $C_i \cap \partial X_{\geq 0}$, which is a subset of the real algebraic variety $C_i(\mathbb{R})$.

For $i = 1, 2, \dots, r$, $(C, C_{i,\geq 0})$ are the *boundary components* of $(X, X_{\geq 0})$. $C_{i,\geq 0}$ is a semi-algebraic set. From axiom (P1), it is required that $C_{i,\geq 0}$ is of dimension $D - 1$.

We now define an orientation on $C_{i,\geq 0}$. $U \subset \mathbb{R}^{D-1} \times \mathbb{R}$ is let be an open set such that $(x, y) \in U \cap (\mathbb{R}^{D-1} \times \mathbb{R}_{\geq 0})$ is a local chart of $X_{\geq 0}$, where $z = 0$ is mapped to the boundary. Euclidean charts may be assumed to be orientated in the regular way.

Assumptions on X

As X is a complex variety, geometric conditions will be imposed on it for the axiom (P2) to have proper meaning. We will let X be a singular complex variety, meaning it contains a singularity. X will be assumed to be a normal variety, meaning that the singular locus will be of co-dimension two or more. If a D -form is rational on X it is also defined to be a D -form rational on the smooth locus X° . If $C \subset X$ is a co-dimension one subvariety, $X^\circ \cap C$ is then open and dense in C , which means $\text{Res}_C \Omega$ is a rational $D - 1$ -form on C .

Axiom (P2) guarantees the following: X must not have holomorphic D -forms, since X did in fact have holomorphic D -forms then $\Omega(X, X_{\geq 0})$ would not be unique. On the other hand, if X does not have nonzero holomorphic D -forms we can see how (P2) can be understood: Say there exists two forms, $\Omega_1(X, X_{\geq 0})$ and $\Omega_2(X, X_{\geq 0})$, which satisfy the aforementioned property, then the difference of both $\Omega_1(X, X_{\geq 0})$ and $\Omega_2(X, X_{\geq 0})$ will have no residues, and is therefore a holomorphic D -form on X , which must vanish at every point. For most of the cases needed, X is a rational variety, which means that X has an open Zariski subset which is isomorphic to an open subset of affine space. We can think of an affine space as a vector space in which an origin does not exist.

2.3.1 Recursive relations of Positive Geometries

- If $D = 1$: If the chart is orientation preserving, a $(+1)$ is assigned to the point $C_{i, \geq 0}$, if not, a (-1) is assigned.
- If $D > 1$: A chart can always be chosen which preserves orientation. The boundary will inherit locally the orientation of the initial $D - 1$ components.

If the orientation of $X_{>0}$ is reversed, this is the same as mapping a positive geometry to another positive geometry.

- For $D = 0$: X is a single point and it is necessary for $X_{\geq 0} = X$. The 0-form $\Omega(X, X_{\geq 0})$ on X , is defined to be ± 1 depending on the orientation of $X_{\geq 0}$.
- For $D > 0$, we must have:

P1 : Every component $(C, C_{\geq 0})$ of $(X, X_{\geq 0})$ is a positive geometry of dimension $D - 1$.

P2 : There exists a unique non-zero rational D -form $\Omega(X, X_{\geq 0})$ on X constrained by the residue relation $\text{Res}_C \Omega(X, X_{\geq 0}) = \Omega(C, C_{\geq 0})$ along every boundary component C , with no other singularities existing elsewhere. $\text{Res}_C(\Omega)$

X is known as the *embedding space*, and D is the dimension of the positive geometry $(X, X_{\geq 0})$. $\Omega(X, X_{\geq 0})$ is what is referred to in the literature as the *Canonical form* of the positive geometry.

2.3.2 Pseudo-Positive Geometries

We will now define a more general type of positive geometry for later use.

Definition Pseudo-Positive Geometry: A pair $(X, X_{\geq 0})$ of the same type of positive geometry, where $X_{\geq 0}$ can now be empty, is a D -dimensional positive geometry, in which the following recursive axioms are satisfied:

- For $D = 0$: X is a singular point. If $X_{\geq 0} = X$, the 0-form $\Omega(X, X_{\geq 0})$ on X is defined to be ± 1 , depending on $X_{\geq 0}$'s orientation. If $X_{\geq 0} = \emptyset$ the 0-form $\Omega(X, X_{\geq 0}) = 0$.
- For $D > 0$: if $X_{\geq 0} = \emptyset$, the 0-form $\Omega(X, X_{\geq 0}) = 0$. If $X_{\geq 0} \neq \emptyset$, it must be that:
 1. Every boundary component $(C, C_{\geq 0})$ of $(X, X_{\geq 0})$ is a pseudo-positive geometry of dimension $D - 1$.
 2. There exists a unique rational D -form $\Omega(X, X_{\geq 0})$ on X which is constrained by the residue relation $\text{Res}_C \Omega(X, X_{\geq 0}) = \Omega(C, C_{\geq 0})$ along every boundary component C , and there exists no singularities elsewhere.

We will note the the residue operator $\text{Res}_C \Omega(X, X_{\geq 0})$, is defined in section 7.1. The same symbols of $X, X_{\geq 0}, \Omega$, as for positive geometries, are used for pseudo-positive geometries. The differences are that for pseudo-positive geometries $X_{\geq 0} = \emptyset$, and we allow $\Omega(X, X_{\geq 0}) = 0$.

The case of a pseudo-positive geometry not being a positive geometry is when $\Omega(X, X_{\geq 0}) \neq 0$. When $\Omega(X, X_{\geq 0}) = 0$, $X_{\geq 0}$ is declared a *null geometry*. An example of a null geometry is the unit circle at the origin.

2.3.3 One-dimensional Positive Geometries

The simplest type of positive geometries are 1-dimensional positive geometries, so we will discuss them now.

If $(X, X_{\geq 0})$ is a zero-dimensional positive geometry, then both X and $X_{\geq 0}$ are points, and $\Omega(X, X_{\geq 0}) = \pm 1$. If $(X, X_{\geq 0})$ is a pseudo-positive geometry instead, then in addition we are allowed to have $X_{\geq 0} = \emptyset$ and $\Omega(X, X_{\geq 0}) = 0$.

Suppose that $\Omega(X, X_{\geq 0}) = 0$ is a one-dimensional pseudo-positive geometry. A genus g projective smooth curve has g independent holomorphic differentials. In the assumptions in A, we know that X must be projective and normal but with no nonzero holomorphic forms, X must have genus 0, and is therefore isomorphic to the line $\mathbb{P}^1$. This means $X_{\geq 0}$ is a closed subset of $\mathbb{P}^1(\mathbb{R}) \cong S^1$. If $X_{\geq 0} = \mathbb{P}^1$, or $X = \emptyset$, then $\Omega(X_{\geq 0})$ and $X_{\geq 0}$ is a pseudo-positive geometry, but not a positive geometry. If these are not satisfied, $X_{\geq 0}$ is a union of closed intervals, and any union of closed intervals is a positive geometry. An example of a generic closed interval is as follows:

Example of One-Dimensional Positive Geometry: An example of such a closed interval is the line segment $[a, b] \subset \mathbb{P}^1$, which is the set of points $\{(1, x) \mid x \in [a, b] \subset \mathbb{P}^1(\mathbb{R})\}$, where $a < b$. The canonical form is given by

$$\Omega([a, b]) = \frac{dx}{x - a} - \frac{dx}{x - b} = \frac{(b - a)}{(b - x)(x - a)} dx. \quad (2.2)$$

Here x is the coordinate of the chart $(1, x) \in \mathbb{P}$, and we orient the line segment in the increasing x direction. We will keep this example in mind for when we see the $n = 4$ kinematic associahedron. The canonical form of a disjoint union of line segments is the sum of the canonical form of the line segments.

2.4 Triangulations of Positive Geometries

As we will see later on, the fatgraph method takes one (partial) triangulation of our positive geometry, i.e. one labelling of a Feynman diagram, and can extract all the necessary information to reconstruct the complete Feynman diagram amplitude. We will now get into the weeds of what a triangulation is, mathematically.

2.4.1 Triangulations of Pseudo-Positive Geometries

We will start by explaining the triangulation of the more general pseudo-positive geometry.

We will start by letting X be an irreducible projective variety, and $X_{\geq 0} \subset X$ being a closed semi-algebraic subset, where $X_{\geq 0}$ is the closure of $X_{>0}$. Here $X_{>0}$ is of the same type described in the assumptions in section 2.3, as in it is an open orientated smooth sub-manifold of $X(\mathbb{R})$, of dimension D . We will let the tuples $(X, X_{i,\geq 0})$, where $i = 1, \dots, r$ be a finite collection of pseudo-positive geometries, which all live in the variety X . We can say that the collection $\{(X, X_{i,\geq 0})\}$ triangulate $(X, X_{\geq 0})$ if the following properties are satisfied:

- Every $\{(X, X_{i,>0})\}$ is contained in $\{(X, X_{i,\geq 0})\}$ and the orientations are the same.
- The interiors $\{(X, X_{i,>0})\}$ of $(X, X_{i,\geq 0})$ are mutually disjoint.
- The union of all $\{(X, X_{i,\geq 0})\}$ gives $\{(X, X_{\geq 0})\}$.

There is a very important feature of canonical forms that we will establish in this section, which is,

- If $\{(X, X_{i,\geq 0})\}$ is a triangulation of $(X, X_{\geq 0})$, then $(X, X_{\geq 0})$ is a pseudo-positive geometry, and the form $\Omega(X, X_{\geq 0})$ is equal to the sum of the forms $\Omega(X, X_{i,\geq 0})$

$$\Omega(X, X_{\geq 0}) = \sum_{i=1}^t \Omega(X, X_{i,\geq 0}). \quad (2.3)$$

2.4.2 Signed Triangulations

There are three ways we can define triangulations, which in the reference material are all referred to as *singed triangulations*, which are; *interior triangulations*, *boundary triangulations*, and *boundary triangulations*.

The collection $\{(X, X_{i,\geq 0})\}$ interior triangulates the empty set, if for every $x \in \bigcup_i (X, X_{i,\geq 0})$, which is not located in $(C, C_{i,\geq 0})$, the amount of sub $(X, X_{i,>0})$ which are positively orientated is the same amount of $(X, X_{i,\geq 0})$ which are negatively orientated,

$$\begin{aligned} & \#\{i | x \in (X, X_{i,>0}) \text{ and } (X, X_{i,>0}) \text{ is positively orientated at } x\} \\ &= \#\{i | x \in (X, X_{i,>0}) \text{ and } (X, X_{i,>0}) \text{ is negatively orientated at } x\}. \end{aligned} \quad (2.4)$$

A boundary triangulation is as follows. For X of dimension D , $\{(X, X_{i,\geq 0})\}$ is said to be a boundary triangulation of the empty set if:

- For $D = 0$, we have $\sum_{i=1}^t \Omega(X, X_{i,\geq 0}) = 0$.
- For $D > 0$, where C is a irreducible subvariety of X , with the dimension $D-1$. As usual $(C, C_{i,\geq 0})$ will be let to be the boundary component of $(X, X_{i,\geq 0})$, along C , however in this instance $C_{i,\geq 0} = 0$. It is required for every C of the collection $\{(C, C_{i,\geq 0})\}$ forms a boundary of the empty set.

The definition of the canonical form triangulation is as follows: $\{(X, X_{i,\geq 0})\}$ is a canonical form triangulation of the empty set, if the sum of the forms of the sub $(X, X_{i,\geq 0})$ s of $(X, X_{\geq 0})$ equal to zero,

$$\sum_{i=1}^t \Omega(X, X_{i,\geq 0}) = 0. \quad (2.5)$$

It is true that:

$$\text{interior triangulation} \implies \text{boundary triangulation} \iff \text{canonical form triangulation}. \quad (2.6)$$

Equation 2.6 shows equation 2.3, as if $\{(X, X_{i,\geq 0})\}$ triangulates $(X, X_{\geq 0})$, then $\{(X, X_{i,\geq 0})\}$ interior triangulates $(X, X_{\geq 0})$. In summary this means that,

$$\text{The canonical form is triangulation independent}. \quad (2.7)$$

2.5 Morphisms of Positive Geometries

Morphisms of positive geometries are important as we will need to change the dimension of our associahedron, and its corresponding differential form, to reduce the form to encode the amplitude.

Now we will define the *push-forward* map of positive geometries. We will consider a surjective meromorphic map $\Phi : M \rightarrow N$, between complex manifolds of dimension d , with ω being defined as a meromorphic top form on M , which means the form is an n -form. Let b be a point in N , and let V be an open subset which contains b . For the map ϕ , being of degree $\deg \phi$, the pre-image $\Phi^{-1}(V)$ is then union of disconnected open subsets U_i , where $i = 1, \dots, \deg \phi$, with $a_i \in U_i$ and $\phi(a_i) = b$. The push-forward is defined to be a meromorphic top form on N :

$$\phi_*(\omega)(b) := \sum_i \psi_i^*(\omega(a_i)), \quad (2.8)$$

where $\psi_i := \phi|_{U_i}^{-1} : V \rightarrow U_i$.

In the language of positive geometries, we can take two pseudo-positive geometries, $(X, X_{\geq 0})$ and $(Y, Y_{\geq 0})$, which are both of dimension D , along with a a morphism,

$$\Phi : X \rightarrow Y, \quad (2.9)$$

which when restricted to the interior of $X_{\geq 0}$, $X_{> 0}$, i.e. $\Phi|_{X_{> 0}} X_{> 0} \rightarrow Y_{> 0}$, is a diffeomorphism which preserves orientation.

If $\Phi : (X, X_{\geq 0}) \rightarrow (Y, Y_{\geq 0})$ along with $\Psi : (Y, Y_{\geq 0}) \rightarrow (Z, Z_{\geq 0})$, are both morphisms, then we can construct a new morphism Π , where $\Pi = \Psi \circ \Phi$. The push-forward is a functorial, which means that it is the story of some sort of functor, i.e. a mapping between categories,

$$\Pi_* = \Psi_* \circ \Phi_*. \quad (2.10)$$

This means that pseudo-positive geometries with morphisms form a category.

An important feature of morphisms of pseudo-geometries is: For a morphism $\Phi : (X, X_{\geq 0}) \rightarrow (Y, Y_{\geq 0})$ of between pairs of positive geometries,

$$\Phi_*(\Omega(X, X_{\geq 0})) = \Omega((Y, Y_{\geq 0})), \quad (2.11)$$

where $\Phi_*(\omega)(b) := \sum_i \Psi_i^*(\omega(a_i))$.

This means that the push forward preserves the canonical form.

Now why is this important? This means that *the push forward is commutative with taking the residue*. As usual $(C, C_{\geq 0})$ will denote the boundary of $(X, X_{\geq 0})$. As in [3], we will take for a given that the map Φ is rational and well defined on $C_{\geq 0}$. From this we can say that $C_{\geq 0}$ will be mapped to $\partial Y_{\geq 0}$. As is stated in more detail, again in [3], C will not contribute to the poles of $\Phi_*(\Omega(X, X_{\geq 0}))$, as Φ collapses on the dimension of C , but if it does not it maps C surjectively onto the D boundary components of Y . When C maps surjectively on $D_{>0}$, which means $(C, C_{\geq 0}) \rightarrow (D, D_{\geq 0})$, is a morphism of pseudo-positive geometries,

$$\text{Res}_D \Phi_*(\Omega(X, X_{\geq 0})) = \Phi_*(\text{Res}_C \Omega(X, X_{\geq 0})) = \Phi_*(\Omega(C, C_{\geq 0})) = \Omega(D, D_{\geq 0}), \quad (2.12)$$

which can be seen to show that,

$$\Phi_*(\Omega(X, X_{\geq 0})) = \Omega((Y, Y_{\geq 0})). \quad (2.13)$$

What all of this means is that if we know the amplitude, *i.e.* the canonical form domain is already known, we can then compute the image of the canonical form, i.e. the associahedron.

2.6 Generalisation of Simplices

For a positive geometry $(X, X_{\geq 0})$, we call $(X, X_{\geq 0})$ a *generalised simplex* if the canonical form $\Omega(X, X_{\geq 0})$, does not contain any zeros within X . The residues of a meromorphic top form which contains no zeros in X , is again a meromorphic top form with no zeros in X . This means that all of the boundary components of $(X, X_{\geq 0})$ are generalised simplices.

2.6.1 The Standard Simplex

The typical example of a generalised simplex is $(\mathbb{P}, \Delta^m)$, which is a positive geometry. $\Delta^m := \mathbb{P}_{\geq 0}^m$, which is the set of points in our real part of our complex projective space $\mathbb{P}^m$ with non-negative coordinates. Δ^m will be our *standard simplex*. the canonical form of our standard simplex is,

$$\Omega(\Delta^m) = \prod_{i=1}^m \frac{d\alpha_i}{\alpha_i} = \prod_{i=1}^m d\log(\alpha_i), \quad (2.14)$$

where our points $(\alpha_0, \alpha_1, \dots, \alpha_m) \in \mathbb{P}^m$, and $\alpha_0 = 1$. This form has a pole which corresponds to the facet where $\alpha_0 \rightarrow 0$, with the (gauge) choice that $\alpha_0 = 1$. We also note that this is very near resembling the form of our “fatgraph curve integral”, where the α_i ’s will be our headlight functions.

2.6.2 A projective Simplex

$(\mathbb{P}^m, \Delta^m)$ is a positive geometry, called a *projective m -simplex*, in $\mathbb{P}^m$ cut out by $m+1$ linear inequalities. Y is now introduced, which is an element of $\mathbb{P}^m$, to denote a point in projective space with homogeneous components Y^I , where $I = 0, 1, \dots, m$. A Linear inequality can be written in the form $Y^I W_I \geq 0$ for some vector $W \in \mathbb{R}^{m+1}$, where W_I , are the components, and I is summed over. As usual we can define $Y \cdot W := Y^I W_I$. The vector W can be labelled a dual vector, with our labellings of Y and W , we can now define a projective simplex.

$$\Delta = \{Y \in \mathbb{P}^m(\mathbb{R}) | Y \cdot W_i \geq 0 \text{ for } i = 1, \dots, m+1\}. \quad (2.15)$$

The inequality is first evaluated in euclidean, before being mapped to projective space. In this case our W_i s are projective dual vectors which correspond to the facets of our simplex. Each of the boundaries of a projective simplex is also a projective simplex. This then means that the boundaries of a projective simplex satisfy the axioms of a positive geometry.

It is important to define the canonical form, $\Omega(\Delta)$, of our projective simplex, Δ in terms of the vertices and the facets of itself. In order to do so, we will let Z_i^I denote the vertices of Δ , for $i = 1, \dots, m+1$. In this case, we will similarly to Y^I , let each Z_i have upper indices I , so we will have Z_i^I .

The canonical form of Δ in term of its vertices:

$$\Omega(\Delta) = \frac{s_m \langle Z_1 Z_2 \dots Z_{m+1} \rangle^m \langle Y d^m Y \rangle}{m! \langle Y Z_1 \dots Z_m \rangle \langle Y Z_2 \dots Z_{m+1} \rangle \dots \langle Y Z_{m+1} \dots Z_{m-1} \rangle}. \quad (2.16)$$

In this case the angle brackets $\langle \dots \rangle$ represent the determinant of the column vectors of what is contained within the brackets. $\langle Y d^m Y \rangle$ is defined in the appendix 7.1. Our $s_m = -1$ when $m = \{1, 5, 9, 13, \dots\}$, and $s_m = 1$ otherwise.

If the W_i facets or adjacent to the vertices $Z_{i+1}, \dots, Z_{i+m}$, $W_i \cdot Z_j = 0$, where $j = i+1, \dots, i+m$. It is then taken as stated in [3].

The canonical form of Δ in term of its vertices:

$$\Omega(\Delta) = \frac{\langle W_1 W_2 \dots W_{m+1} \rangle \langle Y d^m Y \rangle}{m! (Y \cdot W_1) (Y \cdot W_2) \dots (Y \cdot W_{m+1})}. \quad (2.17)$$

The coefficients of $\langle Y d^m Y \rangle$ can be defined to be $[1, 2, \dots, m+1]$, which gives us the following equitation,

$$[1, 2, \dots, m+1] = \underline{\Omega}(\Delta) \quad (2.18)$$

$\underline{\Omega}(\Delta)$ is defined in section 7.1 of the appendix. We can see an example of this in 2.2. This is one of the most basic simplices we can see, which is the line segments of $\mathbb{P}^1(\mathbb{R})$. This can be seen as a segment $[a, b] \in \mathbb{P}$, where its vertices are

$$Z_1^I = (1, a) \quad Z_2^I = (1, b), \quad (2.19)$$

where $(a < b)$.

We will also state without proof, from the appendix of [1], that our formula 2.17 can be rewritten, by applying a $GL(m+1)$ transformation, fixing the W_i 's to the identity matrix, as:

$$\Omega(\Delta) = \frac{\langle W_1 W_2 \dots W_{m+1} \rangle \langle Y d^m Y \rangle}{m! (Y \cdot W_1) (Y \cdot W_2) \dots (Y \cdot W_{m+1})} = \bigwedge_{j=1}^m d \log \left(\frac{Y \cdot W_j}{Y \cdot W_{m+1}} \right). \quad (2.20)$$

2.7 Generalisation of Polytopes

We will now look at more generalised polytopes, whose canonical forms contain zeros, which is what we will link to scattering amplitude poles of our differential forms. Before this we are going to take a brief aside to include some definitions on projective polytopes.

2.7.1 Aside on Projective Polytopes

We will let $V \in \mathbb{R}^{m+1}$. A convex cone is a subset $C \in V$, if it is closed under: (i) addition and (ii) scalar multiplication, in $\mathbb{R}_{\geq 0}$. This means that the zero-vector is always contained in a convex cone. A convex cone is pointed if it does not contain a line.

A polyhedral cone is the span of non-negative vectors

$$\text{Cone} := \text{span}_{\mathbb{R}_{\geq 0}}(Z_1, Z_2, \dots, Z_n) = \left\{ \sum_i a_i Z_i \mid a_i \in \mathbb{R}_{\geq 0} \right\}, \quad (2.21)$$

where Z_i are vectors, and $i = 1, \dots, n$. Something we can point out is that $\dim C$ is equal to dimension of the vector space which C spans.

We will call C a pointed polyhedral cone. If no Z_i vectors can be removed while still spanning C , the vectors Z_i will be called the generators, or edges, of C . The interior of C , is the Cone, which is defined as,

$$\text{Int}(C) := \text{span}_{\mathbb{R}_{> 0}}(Z_1, Z_2, \dots, Z_n) = \left\{ \sum_i a_i Z_i \mid a_i \in \mathbb{R}_{> 0} \right\}. \quad (2.22)$$

A cone C is called *simplicial* if it has $\dim C$ generators. Now we will begin to define facets, which we will use to factorise our associahedron. A face of C , we will call F , is the intersection $F = C \cap H$, where H is a hyperplane, and $H \in V$, where C is completely on one side of H . A face F is called a *facet* of C if $\dim F = \dim C - 1$.

A triangulation of a polyhedral cone C is a collection of cones $C_i, \dots, C_r$, where all sub cones are simplicial and have the same dimension of C , along with the following:

- C is such that $C = \cup_i C_i$.
- The intersection F , of any two sub cones, $F = C_i \cap C_j$ is a face of both C_i and C_j .

A projective polytope $\mathcal{A} = \text{Conv}(Z)$, is the image of a pointed polyhedral cone $C = \text{Cone}(Z)$, in $\mathbb{P}^m$. The facets of $\mathcal{A}$ are defined to be the images of the facets and faces of C , respectively. Something to note is that the interior of a projective polytope $\mathcal{A}$, where $\mathcal{A}$ has an orientation, has the structure of an orientated manifold.

2.7.2 Continuation of projective Polytopes, With Euclidean Polytopes

A positive geometry $(\mathbb{P}^m, \mathcal{A})$, can be formed for every projective geometry $\mathcal{A}$. Looking back to section 2.4, this means every polytope $\mathcal{A}$ can be triangulated by projective simplices. Also by section 2.6.2, every simplex is itself a positive geometry, which is to say that $(\mathbb{P}, \mathcal{A})$ is a positive geometry.

When working with our associahedron, we will be visualising them in euclidean space, so it is important to be able to visualise our polytopes in euclidean space. This means we will have to define a euclidean polytope. To do this we will use the *little group* to “gauge fix” the first component of Z to be 1, i.e. $Z = (1, Z')$, with $Z' \in \mathbb{R}$. The little group has an action on Z which fixes $\mathcal{A}$, which means that $\mathcal{A}$ is left unchanged if every Z_i term is replaced by a positive multiple of itself. Using this background we can then identify our polytope $\mathcal{A} \in \mathbb{P}^m$ with the set,

$$\left\{ \sum_{i=1}^n C_i Z'_i \mid C_i \geq 0, i = 1, \dots, n \text{ and } C_1 + \dots + C_n = 1 \right\}, \quad (2.23)$$

in Euclidean space $\mathbb{R}^m$. A matrix Z can be constructed from the points, $Z_1, \dots, Z_n$, where Z is an $n \times (m+1)$ matrix. This means Z can be thought of as a linear map, $Z : \mathbb{R}^n \rightarrow \mathbb{R}^{m+1}$. Furthermore, our polytope $\mathcal{A}$ is now the image $Z(\Delta^{n-1})$ of the standard $(n-1)$ -dimensional simplex in $P^{n-1}(\mathbb{R})$.

2.7.3 Dual Polytopes

Dual polytopes are important because they connect our kinematic associahedron method with our fatgraph method, and we will see how in chapter 3.

Let $(\mathbb{P}^m, \mathcal{A})$ be a convex polytope, and let $Y \in \mathbb{P}^m(\mathbb{R})$ be a point away from any boundary component of our polytope. Our *dual polytope* $\mathcal{A}_Y^*$ is defined to be a convex polytope within the linear dual of $\mathbb{P}^m$. In the simplest terms, we can see this as a polytope in projective space, whose dual in the dual projective space has swapped its point with lines, and lines with points

Briefly we will work in $\mathbb{R}^{m+1}$ for simplicity, where $Y \in \mathbb{R}^{m+1}$. Each facet of $\mathcal{A}$ is given by the zero set of a dual vector $W \in \mathbb{R}^{m+1}$. Still in euclidean space, the overall sign of W is picked so that $W \cdot Y > 0$, for Y . Now we can say than our facets are orientated positively relative to Y . For $W_1, \dots, W_r$ being facets of $\mathcal{A}$ we can define our dual convex polytope of $\mathcal{A}$ at a point Y away from the boundary as:

$$\mathcal{A}_Y^* := \text{Conv}(W_1, \dots, W_r) := \left\{ \sum_{j=1}^r C_j W_j \in \mathbb{P}^m \mid C_j \geq 0, j = 1, \dots, r \right\}. \quad (2.24)$$

If $Y \in \text{Int}(\mathcal{A})$, we will denote this as $\mathcal{A}^*$, and refer to it simply as the *dual of* $\mathcal{A}$. In this case we can simply define $\mathcal{A}^*$ as,

$$\mathcal{A}^* = \{ W \in \mathbb{P}^m \mid W \cdot Y \geq 0 \text{ for all } Y \in \mathcal{A} \}. \quad (2.25)$$

For any Y , to assign an orientation we must do as follows. If the facets $W_1, \dots, W_r$ are oriented relative to $\text{int}(\mathcal{A})$, and let s denote the amount of times $W_j \cdot Y$ are negative, we will then orient $\mathcal{A}_Y^*$ based on the parity of s . This means that $\mathcal{A}^*$ is positively oriented.

If a polytope $\mathcal{A}_Y^*$ is dual to $\mathcal{A}$, they have opposite combinatorial properties. What this means is that the vertices of $\mathcal{A}$ correspond to the facets of $\mathcal{A}_Y^*$, and vice-versa.

An important note about dual polytopes is, given a signed triangulation $\mathcal{A} = \sum_i \mathcal{A}_i$ of $\mathcal{A}$, where $\mathcal{A}$ is a convex polytope of sub convex polytopes $\mathcal{A}_i$, and a point Y not near a boundary component, we can say that,

$$\mathcal{A} = \sum_i \mathcal{A}_i \implies \mathcal{A}_Y^* = \sum_i \mathcal{A}_{iY}^*. \quad (2.26)$$

This means that,

$$\textit{The dualisation of polytopes "commutes" with triangulations.} \quad (2.27)$$

Some other notes on duals: The volume has a signed nature, what this means is that there is independence of triangulations

$$\mathcal{A}_Y^* = \sum_i \mathcal{A}_{iY}^* \implies \text{Vol}(\mathcal{A}_Y^*) = \sum_i \text{Vol}(\mathcal{A}_{iY}^*), \quad (2.28)$$

Where $\text{Vol}(\mathcal{A}_Y^*)$ is defined as,

$$\underline{\Omega}(\mathcal{A}) = \text{Vol}(\mathcal{A}_Y^*) := \frac{1}{m!} \int_{W \in \mathcal{A}_Y^*} \frac{\langle W d^m W \rangle}{(Y \cdot W)^{m+1}}. \quad (2.29)$$

A proof of this is in section 7 of [3].

We know from section 2.4 the canonical form is triangulation independent, so from what we have already show so far, we can claim that the canonical rational function is triangulation independent:

$$\mathcal{A} = \sum_i \mathcal{A}_i \implies \underline{\Omega}(\mathcal{A}) = \sum_i \underline{\Omega}(\mathcal{A}_i). \quad (2.30)$$

Using equations 2.26, 2.32, 2.30, we can combine them to see that,

$$\text{Vol}(\mathcal{A}_Y^*) = \sum_i \text{Vol}(\mathcal{A}_{iY}^*) = \sum_i \underline{\Omega}(\mathcal{A}_i) = \underline{\Omega}(\mathcal{A}), \quad (2.31)$$

which proves that the volume is signed.

In doing so we have also shown the the canonical form of the polytope $\mathcal{A}$, namely $\underline{\Omega}(\mathcal{A})$, is determined by the volume of the dual polytope, $\text{Vol}(\mathcal{A}^*)$,

$$\underline{\Omega}(\mathcal{A}) = \text{Vol}(\mathcal{A}^*). \quad (2.32)$$

2.8 Simple polytopes

Simple polytopes are just d -dimensional polytopes, where each vertex is adjacent to d faces. In our case each vertex of $\mathcal{A}$ will have $\dim \mathcal{A}$ adjacent edges.

Now we will compute the *canonical form of simple polytopes*, which will given by (almost) the same formula we will see in our associahedron method. The associahedron is a simple polytope, so this section is of vital importance.

We will let $\mathcal{A}$ denote a convex, simple polytope of dimension m . We will first state the formula and then give a short proof by induction.

$$\Omega(\mathcal{A}) = \sum_{\text{Vertex}} \text{sign}(Z) \bigwedge_{\alpha=1}^m d \log(Y \cdot W_\alpha). \quad (2.33)$$

Here the Z vertices have duals W_α 's, which are their corresponding facets, where there are m of them. The $\text{sign}(Z) \in \{\pm 1\}$ tells us the orientation of the facets $W_1, \dots, W_{m-3}$, which is antisymmetric. We can also take note that our associahedron is of dimension $n - 3$, for n particle scattering. This expression would not be well defined for a non-simple polytope.

Now we can see this formula by induction, in m dimensions. For $m = 0$, $\mathcal{A}$ is a singular point, and the canonical form is ± 1 , based on its orientation. In the case that $m > 0$, our claim has been proven for all cases in dimension less than m . The canonical form is uniquely defined by its poles and residues, which means that equation 2.33 is correct to first order. This means it has poles on the facets, as needed. As well as this, for any facet, F , given by $Y \cdot W = 0$, the residue of 2.33 is

$$\sum_{\text{Vertex}} \text{sign}(Z') \bigwedge_{\alpha=1}^{m-1} d \log(Y \cdot W_\alpha), \quad (2.34)$$

where all vertices Z' , adjacent to the facet F are summed over. We can now say, by induction, we have the required canonical form of the facet F , $\Omega(F)$, which completes our derivation.

The form 2.33, will be used to calculate our scattering amplitude.

Chapter 3

The Kinematic Associahedron

We can now move on to our discussion of the associahedron[1]. This method of computing scattering amplitudes will be seen to formulate scattering amplitudes through differential forms, as well as volumes. This will take place in massless $\text{Tr}\phi^3$ theory, as stated in the introduction:

$$\mathcal{L}_{\text{Tr}(\phi^3)} = \text{Tr}(\partial\phi)^2 + g\text{Tr}(\phi^3), \quad (3.1)$$

where ϕ is an $N \times N$ scalar matrix.

3.1 Planar Scattering Forms on Kinematic Space

3.1.1 Kinematic Space

To begin with, kinematic space $\mathcal{K}_n$, for n massless momenta p_i , where $i = 1, \dots, n$, is defined to be the space which the linearly independent Mandelstam variables span, for spacetime dimension $D \geq n - 1$. The Mandelstam variables are defined as follows:

$$s_{ij} := (p_i + p_j)^2 = 2p_i \cdot p_j. \quad (3.2)$$

For $D < n - 1$ we can impose further constraints on the Mandelstam variables, which means there are fewer independent variables. For massless on-shell conditions and momentum conservation, there are n independent constraints,

$$\sum_{j=1, j \neq i}^n s_{ij} = 0, \quad (3.3)$$

for $i = 1, 2, \dots, n$. This means the kinematic space dimensionality is,

$$\dim \mathcal{K}_n = \binom{n}{2} - n = \frac{n(n-3)}{2}. \quad (3.4)$$

In the more general instance, for a set of particle labels $I \subset \{1, \dots, n\}$, the Mandelstam variable can be defined as,

$$s_I := \left(\sum_{i \in I} p_i\right)^2 = \sum_{i, j \in I, i < j} s_{ij}. \quad (3.5)$$

As a result of momentum conservation, where $\bar{I}$ is the capital of I ; $s_I = s_{\bar{I}}$. In the case that $I_1, \dots, I_d$ are mutually disjoint sets, the Mandelstam variable of these sets is

$$s_{I_1, \dots, I_n} := s_{I_1 \cup \dots \cup I_n}. \quad (3.6)$$

For a pair of indices I, J , the Mandelstam variables are defined as follows:

$$s_{I|J} := 2\left(\sum_{i \in I} p_i\right) \cdot \left(\sum_{j \in J} p_j\right) = \sum_{i \in I, j \in J} s_{ij}. \quad (3.7)$$

3.1.2 Planar Kinematic Variables

We will now give a proper definition for kinematic planar variables, of cyclic order. For planar variables, ordered $(1, 2, \dots, n)$:

$$X_{i,j} := s_{i,i+1, \dots, j-1}, \quad (3.8)$$

for the $1 \leq i < j \leq n$. it can be seen that $X_{i,i+1}$ and $X_{i,n}$ vanish. Mandelstam variables can be used to define these planar kinematic variables:

$$s_{ij} = X_{i,j+1} + X_{i+1,j} - X_{i,j} - X_{i+1,j+1}. \quad (3.9)$$

We can check this for the simple 4 point case:

We can use our Mandelstam variables to span our kinematic space. As there are the exact same amount of non-vanishing $\mathcal{K}_n = n(n-3)/2$. For $n = 4$, a possible basis set $\{s := X_{1,3}, t := X_{2,4}\}$. For $n = 5$, $\{s_{12}X_{1,3}, s_{23} = X_{24}, s_{34} = X_{35}, s_{123} = X_{14}, s_{234} = X_{25}\}$.

In a more general sense, for an ordering of $\alpha := (\alpha(1), \dots, \alpha(n))$, of the external particles, α -planar variables can be defined as,

$$X_{\alpha(i), \alpha(j)} := s_{\alpha(i), \alpha(i+1), \dots, \alpha(j-1)}, \quad (3.10)$$

for pairs of $i < j$ modulo n .

It is possible to create a convex n -gon, where each vertex is assigned a momentum, from $(p_1, p_2, \dots, p_n)$. For the $n = 4$ case this will just be the two cuts we can make for the two pairs of diagonal corners. This can be seen in figure 3.2, where the $(1, 2, 3, 4)$ vertex labels represent our (p_1, p_2, p_3, p_4) momenta.

3.1.3 The Planar Scattering Form

Let g denote a cubic tree graph with propagators of the form X_{i_a, j_a} , where $a = 1, \dots, n-3$. For every ordering of our propagators a value of $\text{sign}(g) \in \{\pm 1\}$ is assigned to the graph with the property that each propagator flips the sign. Now we can assign a d log form to the graph as follows,

$$\Omega_{n_g}^{(n-3)} = \text{sign}(g) \bigwedge_{a=1}^{n-3} d \log X_{i_a, j_a}, \quad (3.11)$$

where g represents the specific planar cubic graph, and $\text{sign}(g)$ is resultant of the ordering in the wedge product of the propagators.

We can now introduce the full *planar scattering form* of rank $(n - 3)$,

$$\Omega_n^{(n-3)} := \sum_{\text{planar } g} \text{sign}(g) \bigwedge_{a=1}^{n-3} d \log X_{i_a, j_a}. \quad (3.12)$$

The full planar scattering form sums over all the planar cubic graphs, g . A specific ordering is specified in this construction, so we can denote $\Omega_n^{(n-3)}$, instead as $\Omega^{(n-3)} = [1, \dots, n]$ in the case in which we desire to put emphasis on the ordering. In the case of $n = 3$, $\Omega_n^{(0)} = \pm 1$.

From this we can see there are many possibilities of our scattering form for higher n values. We make the requirement that *the planar scattering form is projective*, which gives a unique choice of scattering form, up to a minus sign.

What this means is there is a requirement of invariance for our scattering form under local $\text{GL}(1)$ transformations, of the form $X_{i,j} \rightarrow \Lambda(X)X_{i,j}$. This is the same as invariance of s_I under $\text{GL}(1)$ transformations, of the form $s_I \rightarrow \Lambda(s)s_I$, for an index set I . This results in the scattering form being fixed up to a minus sign (for our purposes we will disregard the minus sign).

Invariance under local $\text{GL}(1)$ transformations creates a sign-flip rule. If one planar graph, g , can be related by a channel swap of a sub 4 leg graph, to another planar graph g' , we call this a *mutation* from g to g' . For graphs g, g' , with mutated propagators $X_{i,j}, X_{i',j'}$, respectively. Let X_{i_b, j_b} , for $b = 1, \dots, n - 4$ be the propagators shared between both graphs. We can see the Λ dependence of the scattering under a local $\text{GL}(1)$ transformation,

$$(\text{sign}(g) + \text{sign}(g')) d \log \Lambda \wedge \left(\bigwedge_{b=1}^{n-4} d \log X_{i_b, j_b} \right) + \dots, \quad (3.13)$$

which we can see only includes $d \log$ of the propagators which both graphs g and g' share. We evaluate both $\text{sign}(g)$ and $\text{sign}(g')$ at the same propagator ordering, with replacement of $X_{i,j}$ with $X_{i',j'}$. If the dependence of the form on λ disappears, where,

$$\text{sign}(g) = -\text{sign}(g'). \quad (3.14)$$

In the case of each mutation, the form is projective.

This rule of sign flip ensures forms are cyclically invariant up to a sign:

$$i \rightarrow i + 1 \implies \Omega_n^{(n-3)} \rightarrow (-1)^{n-3} \Omega_n^{(n-3)}. \quad (3.15)$$

It takes $(n - 3)$ mutations for a cyclic shift to be achieved, (mod 2).

We need a factorisation of forms, to obtain a factorisation of lower amplitudes, (as will be explained later), which requires this sign flip rule. We only need consider the channel $X_{1,m} \rightarrow 0$, for any $m = 3, \dots, n - 1$ for which,

$$\Omega^{(n-3)}(1, 2, \dots, n) \xrightarrow{X_{1,m} \rightarrow 0} \Omega^{(m-3)}(1, 2, \dots, m - 1, I) \wedge \frac{dX_{1,m}}{X_{1,m}} \Omega^{(n-m-1)}(I^-, m, \dots, n), \quad (3.16)$$

where $p_I = -\sum_{i=1}^{m-1} p_i$ is the sum of all on-shell momenta of internal particles.

If the form only depends on ratios of Mandelstam variables, this is equivalent to the form being projective. In the case of $n = 4$,

$$\Omega^{(1)}(1, 2, 3, 4) = \frac{ds}{s} - \frac{dt}{t} = ds \left(\frac{1}{s} + \frac{1}{t} \right) = d \log \left(\frac{s}{t} \right) = d \log \left(\frac{X_{13}}{X_{24}} \right). \quad (3.17)$$

In the case of $n = 5$,

$$\begin{aligned} \Omega^{(2)}(1, 2, 3, 4, 5) = & d \log X_{1,4} \wedge d \log X_{13} + d \log X_{13} \wedge d \log X_{35} + d \log X_{35} \wedge d \log X_{25} \\ & + d \log X_{25} \wedge d \log X_{24} + d \log X_{24} \wedge d \log X_{14}. \end{aligned} \quad (3.18)$$

It is important to note we can rewrite the amplitude in a projective manner, showing its protective invariance, similar to how a quantum field theory amplitude is Lorentz invariant,

$$\Omega^{(2)}(1, 2, 3, 4, 5) = d \log \frac{X_{13}}{X_{24}} \wedge d \log \frac{X_{13}}{X_{14}} + d \log \frac{X_{13}}{X_{25}} \wedge d \log \frac{X_{35}}{X_{24}}. \quad (3.19)$$

3.2 Kinematic Associahedron

We will now continue on to the main geometric object of this chapter, the *associahedron polytope*.

3.2.1 The Associahedron from Planar Cubic Diagrams

The Associahedron is built from polytopes, so we will briefly discuss some concepts regarding them. A *boundary* of a polytope can be a boundary of any co-dimension of the polytope. A boundary of dimension k is a k -boundary, and a *facet* is a co-dimension 1 boundary. A convex polygon, a polygon with all angles less than 180° , of n -sides, is what we will call an n -gon. We can define a *diagonal* is a straight line which connects any two non-adjacent vertices. A collection of mutually non-intersecting diagonals is *partial triangulation*. A *triangulation*, what we will call a *full triangulation* is partial triangulation with the most number diagonals, which is $n - 3$ diagonals.

The properties of a convex polygon of dimension $n - 3$, for $n \geq 3$, are as follows

- Every n -gon, for every $d = 0, 1, \dots, n - 3$, there exists a one-to-one correspondence
- A boundary, F_1 , of co-dimension d and a co-dimension $d + k$ boundary F_2 are adjacent if and only if the partial triangulation of F_2 can be obtained by adding k diagonals to partial triangulation F_1 .

Triangulations with no diagonals are the corresponding polytope's interiors and the *vertices correspond to full triangulations*. We will need to order the n -gons with cyclic ordering from $1, \dots, n$, where *d -diagonal partial triangulations of the n -gon have a one-to-one correspondence with d -cuts on n -particle planar cubic diagrams*. Our n -gon edges are each associated with a particle momentum, where as the diagonals are kinematic variables.

3.2.2 The Kinematic Associahedron

Our associahedron construction depends on the particle ordering.

A region Δ will be defined to be in kinematic space if the following inequalities are imposed on it,

$$X_{ij} \text{ for all } 1 \leq i < j \leq n. \quad (3.20)$$

The kinematic variables $X_{1,i+1}$ and X_{1n} are 0, so do not impose any conditions. As in the section 3.1.1, we know that the kinematic space dimension is the same number as the non-vanishing planar kinematic variables. This leads to the fact that Δ_n is a simplex which has a facet at infinity. This leads to the problem of the associahedron $\mathcal{A}_n$ ($n-3$), which causes problems for $n > 3$, as the associahedron dimension is lower than the kinematic space. The $\mathcal{H}_n \subset \mathcal{K}_n$ subspace is restricted to $(n-3)$, which is defined by,

$$\text{Let } c_{ij} := X_{ij} + X_{i+1,j+1} - X_{i,j+1} \text{ be a positive constant,} \quad (3.21)$$

for every pair of non-adjacent indices $1 \leq i < j \leq n-1$. We know that from our definition of our Mandelstam variables s_{ij} , that

$$c_{ij} = -s_{ij}. \quad (3.22)$$

The dimension of H_n , from the constraints found is,

$$\dim H_n = \dim \mathcal{K}_n - \frac{(n-2)(n-3)}{2} = n-3. \quad (3.23)$$

We can now define our associahedron, $\mathcal{A}_n := H_n \cap \Delta_n$, as a polytope. This means that the associahedron is a polytope of dimension $(n-3)$.

The boundaries of the associahedron is enforced by positivity conditions on the diagonals X_{ij} , which enables co-dimension 1 boundaries when $X_{ij} \rightarrow 0$. By setting $X_{kl} \rightarrow 0$, for a diagonal from points (k,l) to cross a diagonal from (i,j) , we cannot reach a lower dimensional boundary. For our definition of c_{ij} we can replace (i,j) with (a,b) and sum a and b over $i \leq a < j$ and $k \leq b < l$, this results in,

$$X_{jk} + X_{il} = X_{ik} + X_{jl} - \sum_{\substack{i \leq a < b \\ k \leq b < l}} c_{ab}, \quad (3.24)$$

where $0 \leq i < j < k < l \leq n$. In the case that $X_{ik} = 0$ crosses over $X_{jl} = 0$, this results in,

$$X_{jk} + X_{il} = - \sum_{\substack{i \leq a < b \\ k \leq b < l}} c_{ab}, \quad (3.25)$$

which is not consistent as the right hand side of the equation is negative while the left hand side is not negative. This means that we cannot have crossing diagonals which means that $\mathcal{A}_n$ boundaries are labelled by sets of non crossing diagonals, which are partial triangulations. This is what the associahedron mandates.

For the $n=4$ example, the kinematic space, of variables (s, t, u) , which satisfies $s+t+u=0$, is 2-dimensional. The Kinematic Associahedron is defined as the line $0 < s < -u$, and

$const = u < 0$. The $n = 5$ kinematic space is 5-dimensional, the subspace $H_{n=5}$ is 2-dimensional. The subspace is defined by three constants c_{13}, c_{14}, c_{24} . The subspace can be parametrised by the basis (X_{13}, X_{14}) . The associahedron $\mathcal{A}_n$, which is a pentagon, has edges which are,

$$\begin{aligned} X_{13} &\geq 0, \\ X_{35} &= -X_{14} + c_{14} + c_{13} \geq 0, \\ X_{25} &= -X_{13} + c_{13} + c_{14} \geq 0, \\ X_{24} &= -X_{13}X_{14} + c_{13}, \\ X_{14} &\geq 0. \end{aligned} \tag{3.26}$$

3.3 Connection to Amplitudes in $\text{Tr}(\phi^3)$ Theory.

Recall from the introduction that we have applied a cyclic labelling to our external momenta in our $\text{Tr}(\phi^3)$ in the $1/N$ planar limit ¹.

Throughout this discussion we will be using the mathematical background of section ???. We will be trying to prove two claims that we will make now;

- The pullback of our cyclic scattering form $\Omega_n^{(n-3)}$, to our $(n-3)$ -dimensional subspace $\mathcal{H}_n$, will be the canonical form of our associahedron $\mathcal{A}_n$.
- The canonical form of our associahedron $\mathcal{A}_n$ determines the amplitude of bi-adjoint theory, at tree-level, of identical ordering.

We know that the associahedron is a simple polytope, which has the canonical form being a sum of its vertices. For every vertex Z , we can let X_{i_a, j_a} be its adjacent facets, where $a = 1, \dots, n-3$. We will also let $\text{sign}(Z) \in \{\pm 1\}$, for each facet ordering, denote its orientation, relative to the standard orientation. This allows us to write the canonical form as,

$$\Omega(\mathcal{A}_n) = \sum_{\text{vertex } Z} \text{sign}(Z) \bigwedge_{a=1}^{n-3} d \log X_{i_a, j_a}. \tag{3.27}$$

Here $\text{sign}(Z)$ is evaluated on the ordering of the facets in the wedge product.

What we now want to show is that equation 3.27 is equivalent to the pullback of equation 3.12, in the subspace H_n . This is made easier as there is a direct one-to-one correspondence between vertices Z , and planar cubic graphs g . What does this mean? This means we only have to show that the pullback of our g term is in fact our Z term. We can see this as both planar graphs g , and vertices Z , have the same propagators, X_{i_a, j_a} .

However this is not all, as we notice that in equation 3.27, our $\text{sign}(Z)$ is evaluated on the ordering of the facets in the wedge product, i.e geometrically. On the other hand $\text{sign}(g)$ is evaluated on the specific ordering of the propagators in the wedge product, i.e. $GL(1)$ invariance. What is now required to show they are equivalent is to show that $\text{sign}(Z)$ satisfies that sign flip rule.

¹The source material works in bi-adjoint ϕ^3 , but after our $1/N$ limit and cyclic ordering, we can achieve the same kinematic associahedron structure for $\text{Tr}(\phi^3)$ [7].

Let us begin by showing this equivalence. We can let Z and Z' be vertices who have triangulations related by a mutation. The mutations are defined as relations between planar cubic graphs, but they can also be interpreted from triangulations instead. Two triangulations are related by a mutation if one can be extracted from another due to an exchange of one diagonal. This can be seen as if we have two triangulations of a quadrilateral, each can be obtained from the other by a mutation in the figure below, where we now work with an 8-gon, which nicely demonstrates a triangulation vertex quadrilateral.

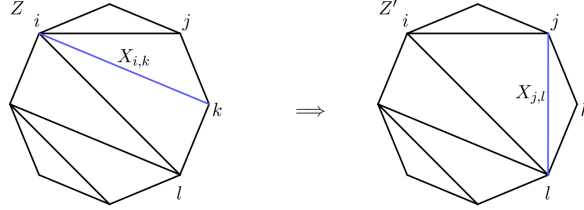


Figure 3.1: Mutation of vertex triangulation [1]

In general, for an n -gon, all we need to do to complete a mutation is to identify a quadrilateral within the triangulation, then exchange the diagonal. This can also be seen for a 4-gon itself, as there are only two possible triangulations.

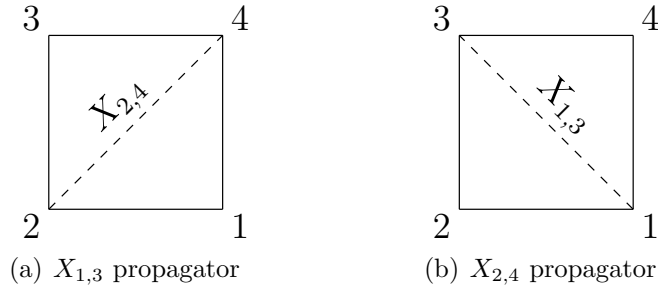


Figure 3.2: Mutation of one propagator gives the other for a quadrilateral.

Now we will take the exterior derivative of equation 3.24, to obtain

$$dX_{j,k} + dX_{i,l} = dX_{i,k} + dX_{j,l}. \quad (3.28)$$

From our 3.1, we can see that on the left hand side of our new equation, while both propagators appear in both diagrams, contrastingly, the propagators on the right hand side are related by a mutation.

Now we can start to equate our two mutations,

$$\bigwedge_{a=1}^{n-3} dX_{i_a, j_a} = - \bigwedge_{a=1}^{n-3} dX_{i'_a, j'_a}. \quad (3.29)$$

We notice a difference of sign, which here gives us the needed sign flip rule:

$$\text{sign}(Z) = -\text{sign}(Z'). \quad (3.30)$$

This means we can now say $\text{sign}(Z) = -\text{sign}(g)$. An important note to make is that

$$\text{sign}(g) \bigwedge_{a=1}^{n-3} dX_{i_a, j_a}, \quad (3.31)$$

does not depend on g , on the pull back. We will also give the following definition,

$$d^{n-3}X := \text{sign}(g) \bigwedge dX_{i_a, j_a}. \quad (3.32)$$

Now we can make a final substitution, of equation 3.32, into equation 3.29, to be left with our final form,

$$\Omega(\mathcal{A}_n) = \left(\sum_{\text{planar } g} \frac{1}{\prod_{a=1}^{n-3} X_{i_a, j_a}} \right) d^{n-3}X. \quad (3.33)$$

We can write this into a slightly neater form,

$$\left(\sum_{\text{planar } g} \frac{1}{\prod_{a=1}^{n-3} X_{i_a, j_a}} \right) d^{n-3}X = m_n d^{n-3}X, \quad (3.34)$$

where we identified that what is within the brackets on the left hand side, is equal to m_n , the expected amplitude.

3.3.1 Examples

We will now finally show, how through our complete formalism, we can calculate particle amplitudes.

We will start with the simplest $n = 4$ case. We will define our Mandelstam variables as follows, in line with the source material,

$$\begin{aligned} s &:= X_{1,3}, \\ t &:= X_{2,4}, \\ u &:= -X_{1,3} - X_{2,4} = -c_{13}. \end{aligned} \quad (3.35)$$

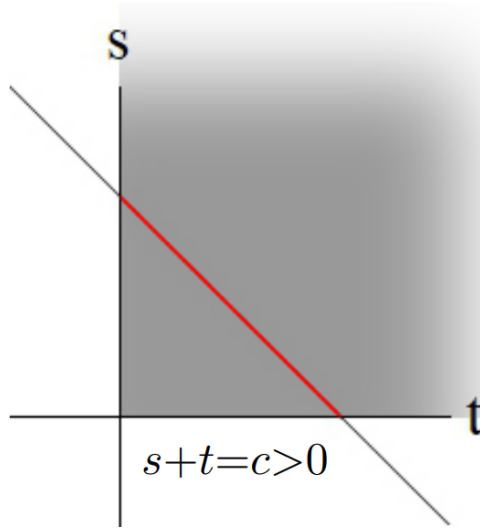
Here we can see that u is a negative constant. We are left with figure 3.3, which is just a line segment. The canonical form of the shape is

$$\Omega(\mathcal{A}_{n=4}) = \left(\frac{1}{s} - \frac{1}{s+u} \right) ds = \left(\frac{1}{s} + \frac{1}{t} \right) ds. \quad (3.36)$$

Here we used the fact that the line segment is simply $t = -s - u$. We are left with $m_n ds$. We will now pull back our form to our subspace H_n . As $u = \text{const}$ on $H_{n=4}$, and $s + t + u = 0$. This means $dt = -ds$, we again are left with,

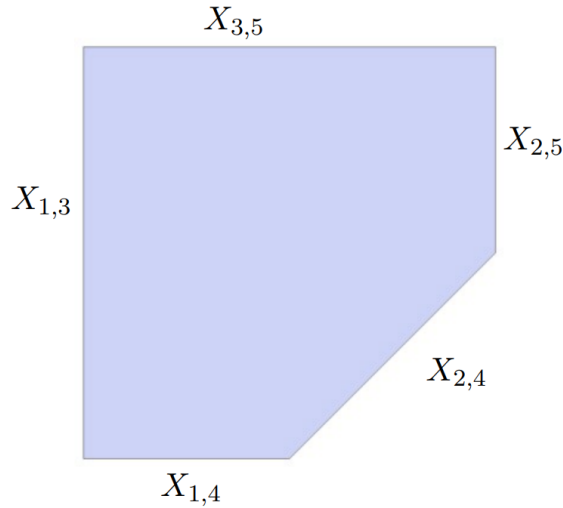
$$\Omega_{n=4}^{(1)}|_{H_{n=4}} = \left(\frac{1}{s} + \frac{1}{t} \right) ds, \quad (3.37)$$

and we see that both equation are equal.


 Figure 3.3: $n = 4$ associahedron [1].

Now we will move onto the more complicated example, $n = 5$. We have our associahedron defined with the kinematic variables in equation 3.26, seen in figure 3.4. In order to be able to use equation 3.32, in our canonical form 3.18, we need to show that $dX_{1,4}dX_{1,3} = dX_{1,3}dX_{3,5} = dX_{1,4}dX_{2,4} = dX_{2,5}dX_{3,5} = dX_{2,4} = dX_{2,4}$. We will do this, for the first identity through identifying $X_{3,5} = -X_{14} + c_{14} + c_{13}$, then using the Jacobi determinant for change of variables we get $\partial(X_{1,4}, X_{1,3})/\partial(X_{1,3}, X_{3,5}) = 1$. This can be similarly achieved for the other equalities. This then gives us the canonical form on the pull back of,

$$\Omega_{n=5}^{(2)}|_{H_{n=5}} = \left(\frac{1}{X_{1,3}X_{1,4}} + \frac{1}{X_{3,5}X_{1,3}} + \frac{1}{X_{1,4}X_{2,4}} + \frac{1}{X_{2,5}X_{3,5}} + \frac{1}{X_{2,4}X_{2,5}} \right). \quad (3.38)$$


 Figure 3.4: $n = 5$ associahedron [1].

3.4 Factorisation

This section is one of the most important in the whole work. In the introduction we stated that we wanted to see manifest in our geometric interpretation of scattering amplitudes, locality and unitarity. We said earlier that locality was able to be seen through logarithmic poles at boundaries of our kinematic associahedron. Now we want to show unitarity, which we argued was shown in the QFT method via factorisation, now we will show factorisation, and inherently unitarity, through factorising our polytopes into smaller polytopes.

An important note to make is that the associahedron *factorises combinatorically*. This means for a facet F which corresponds to a diagonal, dividing an n -gon into an m -gon and an $(n - m + 2)$ -gon. The sub polygons give canonical properties for two lower dimensional associahedron, $\mathcal{A}_m$ and $\mathcal{A}_{(n-m+2)}$, As well as this the facet is identical combinatorially,

$$F \cong \mathcal{A}_m \times \mathcal{A}_{(n-m+2)}, \quad (3.39)$$

where F is the facet of the associahedron $\mathcal{A}_n$.

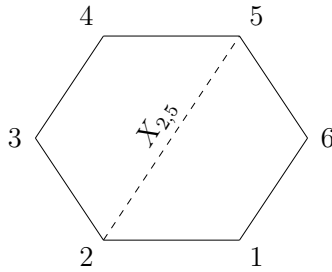


Figure 3.5: Diagonal dividing 6-gon into to sub-polygons.

In the above figure, the product of the two associahedra, which correspond to both sub-polygons, are seen to be combinatorically identical to the facet, corresponding to the dividing diagonal, on the original associahedron.

$$F(X_{4,7}) \cong \mathcal{A}_{n=4} \times \mathcal{A}_{n=4}. \quad (3.40)$$

This can be checked in reference to figure 7.1 in our appendix, which shows are $n = 6$ associahedra, as the product of the two lower associahedra is a product of line segments, so we are left with our facet being a quadrilateral. The reason why we have strayed away from our $n = 4$ and $n = 5$ corresponding polygons for this example, as the sub-polygons need to correspond to at least $n = 4$ associahedra.

The associahedron is a simple polytope, where each of its vertices is adjacent to precisely $\dim \mathcal{A}_n$ facets. in terms of the associahedron, each vertex with its corresponding triangulation has adjacent facets for $(n - 3)$ diagonals.

To explain this in a more concrete manner, we will talk about two different types of factorisation: *Geometric Factorisation*, and *Amplitude Factorisation*. In order to do this we will write our kinematic associahedron for n particle scattering as, $\mathcal{A}(1, 2, \dots, \bar{n})$, in order to show the importance of the specific ordering of the particles. The $\bar{n}$ specifies the ordering of our subspace H_n , and from equations 3.23 and 3.21, we can see it is not dependent on adjacent indices. Now we will explain the two types of factorisation:

Geometric Factorisation: Constraining the associahedron to when $X_{i,j} = 0$, is combinatorically equal to the product of sub polytopes,

$$\mathcal{A}_n|_{X_{i,j}=0} \cong \mathcal{A}_L \times \mathcal{A}_R. \quad (3.41)$$

where $\mathcal{A}_L$ and $\mathcal{A}_R$ are defined as follows,

$$\begin{aligned} \mathcal{A}_L &:= \mathcal{A}(i, i+1, \dots, j-1, \bar{I}) \\ \mathcal{A}_R &:= \mathcal{A}(1, \dots, i-1, I, j, j+1, \dots, \bar{n}). \end{aligned} \quad (3.42)$$

The overline again just specifies the specific ordering.

Amplitude Factorisation: The residue of our canonical form, along the facet $X_{i,j}$ factors as,

$$\text{Res}_{X_{i,j}=0} \Omega(\mathcal{A}_n) = \Omega(\mathcal{A}_L) \wedge \Omega(\mathcal{A}_R) = \Omega(\mathcal{A}_L \times \mathcal{A}_R), \quad (3.43)$$

where in the last step we have used the identity provided in appendix A.1 of [6].

We will now prove both, starting with geometric factorisation. The two sub associahedrons can be constructed by two independent associahedra, within their own unique kinematic spaces. In this case their labels of indices are just for ease of use. This can be seen for two planar variables, which we will call $\mathcal{A}_L$ and $\mathcal{A}_R$:

$$\begin{aligned} \mathcal{A}_L &: L_{a,b} \text{ for } i \leq a < b < j, \\ \mathcal{A}_R &: R_{a,b} \text{ for } 1 \leq a < b < n, \text{ except } i \leq a < b < j. \end{aligned} \quad (3.44)$$

We can visualise our planar variable indices in figure 3.5. As we did for our regular kinematic associahedra, we will need to pick a kinematic basis for $\mathcal{A}_L$ and $\mathcal{A}_R$, which will be $L_{a,b}$ and $R_{a,b}$ respectively. Each basis will be correspondent to a particular triangulation of each of their respective associated sub polygons. Now we can say that we have a basis for our subspace $H_n|_{X_{i,j}=0}$, as each diagonal corresponds to a specific planar variable. We will call these constants l_{ab} and r_{ab} , for the left and right associahedra respectively. We will now make some assumptions, that $c_{ab} = l_{ab}$ for every l_{ab} , and similarly, $c_{ab} = r_{ab}$ for $a, b \neq I$. As before, I here denotes the intermediary particle.

What we want to do now is to construct a map of the product of our two new associahedra to the subspace constructed by their dividing kinematic variable, $\mathcal{A}_L \times \mathcal{A}_R \rightarrow H_n|_{X_{i,j}=0}$. We do this by equating basis variables, as follows:

$$\begin{aligned} X_{a,b} &= L_{a,b} \text{ for all } L_{a,b}, \\ X_{a,b} &= R_{a,b} \text{ for all } R_{a,b}. \end{aligned} \quad (3.45)$$

We want to be able to write $L_{a,b}$ in terms of basis variables and non-adjacent constants, like how we did in 3.24, and since this holds for $X_{a,b}$, and our assumption on the matching constants, we can do just this. This similarly also holds for our $R_{a,b}$.

To complete our proof of geometric factorisation, we need to show that the image of the embedding itself is contained within the facet $\mathcal{A}_n|_{X_{i,j}=0}$, through showing that all planar $X_{a,b}$ propagators are positive in that embedding, except $X_{i,j} = 0$. If a propagator's diagonals do not cross the $X_{i,j}$ diagonal, this is seen to be true as $X_{a,b} = R_{a,b}$ or $X_{a,b} = L_{a,b}$. What

about the case of a propagator $X_{k,l}$, whose diagonal crosses the diagonal of $X_{i,j}$, where $1 \leq i < k < j < l \leq n$? We can now reformulate equation 3.24, where j, k are swapped, and $X_{i,j} = 0$. This leaves us with,

$$X_{k,l} = X_{k,j} + X_{i,l} + \sum_{\substack{i \leq a < k \\ j \leq b < l}} c_{ab}. \quad (3.46)$$

As $A < k < j \leq b$, and $i \leq a \leq b < j$, we know that $X_{k,j}$, is a diagonal within the left sub polygon, and $X_{i,l}$ is a diagonal of the right polygon, and they are both positive. As every term on the right hand side are all individually positive, then the $X_{k,l}$ propagator is positive, as needed. We can see from this that $X_{k,l}$ is strictly positive, which means it cannot be cut. This then implies it does not violate the planar graph structure required for the associahedron formulation to exist. We can now see there is a one-to-one map between the sub associahedra, and the subdividing facet.

We have now proved geometric factorisation, leaving all that remains to prove is the amplitude factorisation, which will manifest unitarity.

Using 2.12, we can say that,

$$\text{Res}_{X_{i,j}=0} \Omega(\mathcal{A}_n) = \Omega(\mathcal{A}_n|_{X_{i,j}=0}) = \Omega(\mathcal{A}_L \times \mathcal{A}_R) = \Omega(\mathcal{A}_L) \wedge \Omega(\mathcal{A}_R). \quad (3.47)$$

The second relation can be seen from 3.41, and the third from 7.12. Now we can say that as our amplitudes factor, and quantum field theory amplitudes factor, our geometric interpretation encodes unitarity, as from section 1.2.

Chapter 4

Fatgraph Method

4.1 New Formulation

For the new formulation of calculating scattering amplitudes we will be working in $\text{Tr}(\phi^3)$ theory. As mentioned previously, this means we will be working with scalar particles, meaning we do not have to consider spin or angular momentum, and in regards to possible Feynman diagrams, each vertex must be the connecting point of three edges, i.e. a cubic vertex. We will also be working in the $1/N$ limit, allowing for cyclic labelling of external momenta, and now we will also consider the massive case. As many steps that will be discussed in this method which are either new ideas entirely or reformulations of old ideas I will start by giving an example of how this new method formulates itself. The steps that will be completed are as follows: (i) “fatten” one of our labelled Feynman diagrams, (ii) draw all possible *non-boundary* curves, (iii), construct a *mountainscape diagram* for each curve, (iv) for each mountainscape-diagram construct the *F-polynomial* and **g**-vector, (v) from each *F*-polynomial construct the *curve matrix*, (vi) from each curve matrix construct the *u-variable*, from each u-variable construct the *headlight-function*, (vii) from the **g**-vectors plot the *curve space*, (ix) use the plot to set bounds for the integral over headlight functions, and finally (x), compute the integral.

A note on notation in this section, we will drop the comma in between the i and j subscripts in our kinematic variables and $\mathcal{A}_n$ will denote the amplitude of n particle scattering. Also we will be working in massive $\text{Tr}(\phi)^3$ theory,

$$\mathcal{L} = \text{Tr}(\partial\phi)^2 + m^2\text{Tr}(\phi^3) + g\text{Tr}(\phi^3). \quad (4.1)$$

4.2 Example $n = 4$

4.2.1 $n = 4$ Fatgraph

This is an original example, not found in the source material. In our new formulation the first and most important step is to “fatten” our regular Feynman diagram, resulting in a fatgraph. The choice of which of the two labelled diagrams in the figure 1.2 is arbitrary, as it will lead to the same result, up to a relabelling. We can see how this is done in the figure below.

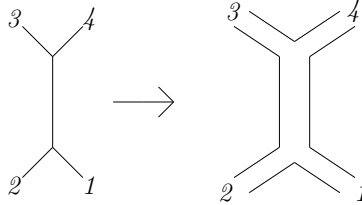


Figure 4.1: Feynman graph fattening.

In doing so this enable us to draw curves through said fatgraph. These curves, as will be shown, encode all the information necessary to perform our amplitude calculations. To start we will begin by stating that this new “fattened” Feynman diagram, has two independent non-boundary curves. We will see by the end of this example, how each curve encodes the information of each unique labelling of the Feynman tree diagram. The following example is not in the source paper, but can be used to build up foundations of the new curve integral method in as simple a manner as possible.

4.2.2 C_{13} Curve

The first step now, after fattening our graph, is to draw a non-boundary curve through it. A non-boundary curve is simply a curve that changes direction at least once from start to finish. The first curve we will draw is seen in the figure below.

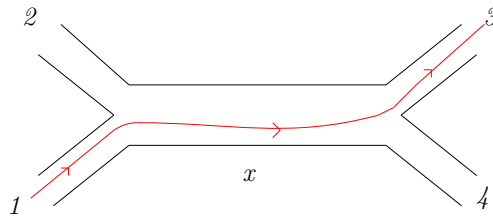


Figure 4.2: C_{13} (non-boundary) curve of four-leg tree fatgraph.

We can see in the above figure that we have a curve, which we will label the C_{13} curve, which goes from external edge 1, through internal edge x , to external edge 3. We will also say that C_{13} turns right from 1 onto x , and then left from x onto 3. As this curve turns right, then left it is a non-boundary curve. One can think of this as that this curve changes what internal boundary of internal edge x it is closest to; from the lower to the upper, after passing from external edge 1, to external edge 3. An example of a boundary curve would be a curve from

2, through x , to 3. We want to be able to describe the path this curve has taken now without any reference to the actual fatgraph. In order to do this we will construct something called a mountainscape diagram. For a mountainscape diagram describing a curve, in this case the C_{13} curve, we draw an arrow down from the point if the curve takes a right turn, and down from a point if the curve takes a left turn, as seen in figure 4.3.

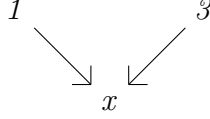


Figure 4.3: Mountainscape diagram of C_{13} curve.

This mountainscape diagram shows a right turn from 1 to x , and a left turn from x to 3.

4.2.3 g-vectors

From the mountainscape we can identify “peaks” and “valleys”, which are associated with the highest and lowest internal edge in our mountainscape, respectively. In figure 4.3 we identify a single valley at x , and no peak. Each peak is labelled as a 1, and each valley is labelled as a -1 in our $\mathbf{g}$ -vector for said curve. The $\mathbf{g}$ -vector for the C_{13} curve is

$$\mathbf{g}_{C_{13}} = [-1]. \quad (4.2)$$

The plot of $\mathbf{g}$ -vectors gives a “ $\mathbf{g}$ -vector” fan which gives a picture of all the Feynman diagrams which live on the *curve space*, which is an E -dimensional vector space, where E is the number of internal edges of associated tree diagrams. This curve space is what we require perform our integration over to calculate our amplitude.

4.2.4 Sub-mountainscape Diagrams

Also from the mountainscape diagram, we can obtain the sub-mountainscape diagrams. For every mountainscape diagram we start with the null sub-mountainscape diagram, then we begin to build up from the bottom subset of the mountainscape diagram. We choose the lowest subset on the diagram, which is x , and that will be its own sub-mountainscape diagram. We then choose terms that go down into x , such as 1 and 3. Each of the individual subsets of 1 to x and 3 to x will be its own sub-diagram. Then finally, we will include the initial, complete mountainscape diagram.

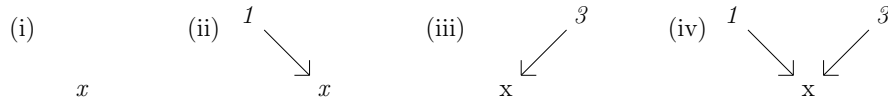


Figure 4.4: Sub-mountainscape diagrams of C_{13} mountainscape (excluding the null).

The mountainscape diagram enables us to create what is called an F-Polynomial. This F-polynomial is a sum of subsets, where each subset is associated with one of the sub-mountainscape-diagrams, along with a 1 subset which is associated with the null-sub-mountainscape diagram. this leaves us with the following F-polynomial:

$$F_{C_{13}} = 1 + x + ax + bx + abx. \quad (4.3)$$

What we have done here is change our 1 subset on the mountainscape for a , and 2 for b , for the purpose of avoiding confusion with our actual polynomial subset of 1. Our a and b subsets are nothing more than connecting subsets, which shows what x subset is connected with our first or last point of our curve. Now we will construct our curve matrix, which has the following structure:

$$M_C = \begin{pmatrix} F_{--} & F_{-+} \\ F_{+-} & F_{++} \end{pmatrix}. \quad (4.4)$$

The F_{--} term includes every subset in our F polynomial which does not include the first or last external edge of the curve, in this case 1 and 2, which in our F polynomial is labelled as a and b , respectively. F_{-+} includes subsets in our F polynomial which include the last but not the first external edge of the curve, and so on. This means the curve matrix for the C_{13} curve of the four-leg tree fatgraph is as follows,

$$M_{C_{13}} = \begin{pmatrix} 1+x & x \\ x & x \end{pmatrix}. \quad (4.5)$$

After constructing our mountainscape diagram, we will use this to construct the u -variable, which is the ratio of the products of the diagonal and off-diagonal terms of our matrix.

$$u = \frac{F_{-+}F_{+-}}{F_{--}F_{++}}. \quad (4.6)$$

So for the C_{13} curve, our u -variable¹ which is defined as the ratio of off-diagonal and diagonal terms of the curve-matrix is as follows:

$$u = \frac{1+x}{x}. \quad (4.7)$$

Aside - Matrix Form of F -Polynomial

We can imagine a mountainscape diagram where the leftmost subset is Y and the second leftmost subset is y . We can then write $F = F_- + F_+$, where terms are grouped whether they are coupled with Y (F_+), or not (F_-). This can also be done for whether terms are coupled to y , with f_+ or f_- , etc.

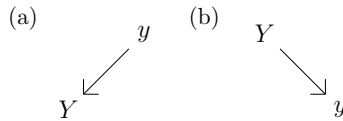


Figure 4.5: Curve matrix explainer mountainscape.

If the Y term is below y , as in (a) in figure 4.5, then we can include Y by itself, as a sub-mountainscape, but if we include y we must include Y , i.e.

$$\begin{aligned} F_- &= f_-, \\ F_+ &= Y(f_+ + f_-). \end{aligned} \quad (4.8)$$

¹Our u -variables are more general versions of u -variables defined on “binary” geometries of Dynkin diagrams.[8]

We can write this in matrix form as

$$\begin{pmatrix} F_- \\ F_+ \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ Y & Y \end{pmatrix}}_{M_L(Y)} \begin{pmatrix} f_- \\ f_+ \end{pmatrix} = \begin{pmatrix} f_- \\ Y(f_+ + f_-) \end{pmatrix}. \quad (4.9)$$

Here $M_L(Y)$ shows a left turn was taken from Y . Now we will consider if instead of going from Y up to y , our mountainscape goes from Y down to y , as seen in mountainscape (b) in figure 4.5. If we have the Y term, we need the y term as well, but if we have the y term, we can have it coupled with and without the Y term, i.e.

$$\begin{aligned} F_- &= f_- + f_+, \\ F_+ &= Y(f_+), \end{aligned} \quad (4.10)$$

which holds the following form,

$$\begin{pmatrix} F_- \\ F_+ \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 0 & Y \end{pmatrix}}_{M_R(Y)} \begin{pmatrix} f_- \\ f_+ \end{pmatrix} = \begin{pmatrix} f_- + f_+ \\ Yf_- \end{pmatrix}. \quad (4.11)$$

$M_R(Y)$ indicates a right turn taken from Y . With both $M_L(Y)$, and $M_R(Y)$ now defined, we can now apply this to any mountainscape-diagram we wish. If we take the mountainscape diagram in figure 4.3, and apply these matrices, we get the result of:

$$M_{C_{13}} = M_R(1) \cdot M_R(x) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ x & x \end{pmatrix} = \begin{pmatrix} 1+x & x \\ x & x \end{pmatrix}. \quad (4.12)$$

We can see this is the same result as achieved in equation 4.5. For each external edge, when filling in the left or right turn matrix, the matrix value will be set to be equal to 1. It is also noted that the determinants $\det M_L(y_i) = \det M_R(y_i) = y_i$, where y_i is an arbitrary edge. This means for a curve matrix,

$$M_C = M_{\text{turn}_1}(y_1) \cdot M_{\text{turn}_2}(y_2) \cdots M_{\text{turn}_{n-1}}(y_{n-2}) \cdot M_{\text{turn}_{n-1}}(y_{n-1}), \quad (4.13)$$

has the following determinant:

$$\det M_C = \prod_{i=2}^{n-1} y_i. \quad (4.14)$$

This can be checked for $M_{C_{13}}$,

$$\det M_{C_{13}} = x, \quad (4.15)$$

as required.

$F(C)$ can be written as explained before in terms of those terms which are directly connected to the first or last edge, y_1 and y_n respectively. This is written as,

$$F(C) = F_{--} + F_{-+} + F_{+-} + F_{++}, \quad (4.16)$$

and from equation 4.4, we know that we can write our curve matrix as ,

$$M_C = \begin{pmatrix} F_{--} & F_{-+} \\ F_{+-} & F_{++} \end{pmatrix}. \quad (4.17)$$

Combining both equations 4.14 and 4.17, we find,

$$F_{--}F_{++} - F_{-+}F_{+-} = \prod_{i=2}^{n-1} y_i. \quad (4.18)$$

Every curve has an associated u -variable.

$$u_C = \frac{F_{-+}F_{+-}}{F_{--}F_{++}} = \frac{(M_C)_{12}(M_C)_{21}}{(M_C)_{11}(M_C)_{22}}. \quad (4.19)$$

In the region we are interested in, $y_i \geq 0$, equation 4.18 ensures that $0 \leq u_C \leq 1$.

4.2.5 The Headlight Function

We will now define a headlight function $\alpha_{C_{13}}$ for the curve C_{13} .

Headlight Function

A general headlight function α_C for a curve C , with $\mathbf{g}$ -vector, $\mathbf{g}_C$ will “light-up” the curve space in the direction of the $\mathbf{g}$ -vector, and vanish in regards to any other direction,

$$\alpha_C(\mathbf{g}_D) = \delta_{C,D} \quad (4.20)$$

The $\mathbf{g}$ -vector space is cut via by cones, bounded by our $\mathbf{g}$ -vectors. From equation 4.20, we know that α_C vanishes when not in a cone cut out by our $\mathbf{g}$ -vector. An explicit expression for our general headlight function can be given in terms of the boundary $\mathbf{g}$ -vectors of a cone, $\mathbf{g}_C, \mathbf{g}_D, \dots, \mathbf{g}_{D_{E-1}}$, and a unique linear function $\mathbf{t}$, evaluated to 1 at $\mathbf{g}_C$ and 0 everywhere else.

$$\alpha_C = \frac{\langle \mathbf{t} \mathbf{g}_D \cdots \mathbf{g}_{D_{E-1}} \rangle}{\langle \mathbf{g}_C \mathbf{g}_D \cdots \mathbf{g}_{D_{E-1}} \rangle}, \quad (4.21)$$

where $\langle \dots \rangle$ denotes the determinant. We now want to define an action, S , which is equal to the sum of the product of each of our curve headlight functions and kinematic variables.

$$S(\mathbf{t}) = \sum_C \alpha_C(\mathbf{t}) X_C, \quad (4.22)$$

recalling the kinematic variable definition $X_{ij} = (p_i + \dots + p_{j-1})^2 + m^2$. For a restriction of $S(\mathbf{t})$ to a single cone bounded by $\mathbf{g}$ -vectors, $\mathbf{g}_{C_1}, \mathbf{g}_{C_2}, \dots, \mathbf{g}_{C_E}$, the only headlight functions, α_{C_S} , which are non-zero are the corresponding $\alpha_{C_1}, \alpha_{C_2}, \dots, \alpha_{C_E}$. As $S(\mathbf{t})$ is linear in said cone, we can parametrise it as $\mathbf{t} = \rho_1 \mathbf{g}_{C_1} + \dots + \rho_E \mathbf{g}_{C_E}$, with $p_i \geq 0$. This enables us to integrate $\exp(-S)$ within said cone, resulting in the standard Schwinger parametrisation of a single Feynman diagram,

$$\int_{\text{cone}} d^E t e^{-S} = \int_0^\infty d^E \rho |\langle \mathbf{g}_{C_1}, \mathbf{g}_{C_2}, \dots, \mathbf{g}_{C_E} \rangle| \prod_{i=1}^E e^{-\rho_i X_C} = \prod_i^E \frac{1}{P_{C_i}^2 + m^2} = m_n. \quad (4.23)$$

The reason for the exponentiation will become clear from the tropicalisation definition. m_n here is used to denote the expected amplitude, as not to confuse it with the mass m . We also note that $\dim E = n - 3$, which is the same dimension as our subspace H_n from chapter 3.

4.2.6 Aside on Tropicalisation

The goal of tropicalisation is to see how a function acts asymptotically, in x -space. This is done by parameterising $x_i \geq 0$ as a function of t_i , where $y_i = e^{(t_i)}$. For a function $u(y_1, y_2) = 1 + y_1 + y_1 y_2 = 1 + e^{t_1} + e^{t_1+t_2}$. When we let $t_i \rightarrow 0$, we end up with $u \rightarrow e^{\max(0, y_1, y_1+y_2)}$. The definition of tropicalisation is such that as $u(t_i) \rightarrow \infty$,

$$u(t_i) \rightarrow e^{\text{Trop}(u(t_i))}. \quad (4.24)$$

In our case, the tropicalisation of our u -variable is our headlight function,

$$\alpha_C = -\text{Trop}(U_C). \quad (4.25)$$

Here the minus sign is just for convention, as it works with our integration bounds. It is because of this parametrisation that we perform the integration over the exponential of our headlight function. After tropicalising our u -variable we acquire our headlight function:

$$\alpha_{C_{13}} = \max(0, x) - x. \quad (4.26)$$

4.2.7 The C_{24} curve

Now we will do the same for the other curve of the four-leg tree fatgraph.

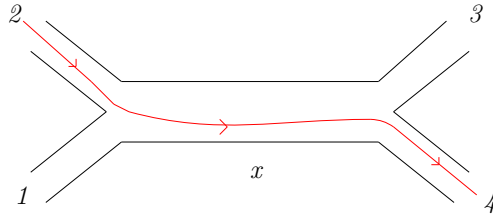


Figure 4.6: C_{24} curve for four-leg tree fatgraph.

As from the rules before, since the curve takes a left turn from 2 onto x , and a right turn from x onto 4 we end up with the following mountainscape diagram:

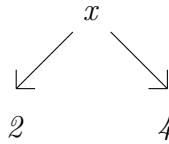


Figure 4.7: Mountainscape diagram for C_{24} curve.

With this we can now calculate our F polynomial,

$$F_{C_{24}} = 1 + a + b + ab + abx, \quad (4.27)$$

and our curve matrix,

$$M_{C_{24}} = \begin{pmatrix} 1 & 1 \\ 1 & 1+x \end{pmatrix}, \quad (4.28)$$

which gives the u -variable and headlight function of

$$u_{C_{24}} = \frac{1}{1+x}, \quad (4.29)$$

$$\alpha_{C_{24}} = \max(0, x). \quad (4.30)$$

Curve Kinematic Variables

Each of our curves also have prescribed to them specific kinematic variables, as seen in equation 1.1. The p terms are the momenta associated with each external edge. This defines the kinematic variables for both the C_{13} and C_{24} as

$$X_{13} = (p_1 + p_2)^2 + m^2, \quad (4.31)$$

$$X_{24} = (p_2 + p_3)^2 + m^2. \quad (4.32)$$

From this we can see that the two kinematic variables associated with each curve has a corresponding kinematic variable associated with one of the four-leg Feynman diagrams.

The Integral

Now we have all the necessary formulae to perform our integral for the amplitude. Our headlight functions ensure that the kinematic variables are only lit up in the space required.

$$\mathcal{A} = \int_{-\infty}^{\infty} dx Z, \quad (4.33)$$

where $-\log(Z) = X_{13}\alpha_{C_{13}} + X_{24}\alpha_{C_{24}}$. This implies

$$\begin{aligned} \mathcal{A} &= \int_{-\infty}^{\infty} dx \exp(-X_{13}\alpha_{C_{13}} - X_{24}\alpha_{C_{24}}), \\ &= \int_{-\infty}^{\infty} dx \exp(X_{13}(x - \max(0, x)) - X_{24}(\max(0, x))), \\ &= \int_{-\infty}^0 dx \exp(X_{13}(x - \max(0, x)) - X_{24}(\max(0, x))) + \\ &\quad + \int_0^{\infty} dx \exp(X_{13}(x - \max(0, x)) - X_{24}(\max(0, x))), \\ &= \int_{-\infty}^0 dx \exp(X_{13}(x)) + \int_0^{\infty} dx \exp(X_{24}(-x)), \\ &= \left[\frac{1}{X_{13}} (\exp(X_{13}(0)) - \exp(X_{13}(-\infty))) \right] + \left[-\frac{1}{X_{24}} (\exp(X_{13}(-\infty)) - \exp(X_{13}(0))) \right], \\ &= \frac{1}{X_{13}} + \frac{1}{X_{24}}. \end{aligned} \quad (4.34)$$

This we can see is the same result as was obtained in equation 1.4.

4.3 Example $n = 5$

The first step again is to fatten one of our labelled five-leg Feynman diagrams, we choose the labelling of (a) in figure 1.4. What we notice here is that we now have two internal edges to label, which we can call (x) and (y) . This example is different to what is contained in the literature, as it considers momenta coming in from left turns, instead of right as described in the source material.

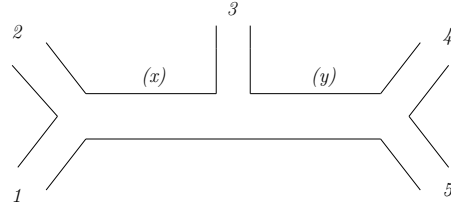


Figure 4.8: Labelled five-leg tree fatgraph

Then we can draw, in this case, our five non-boundary curves, as seen in figure 4.9.

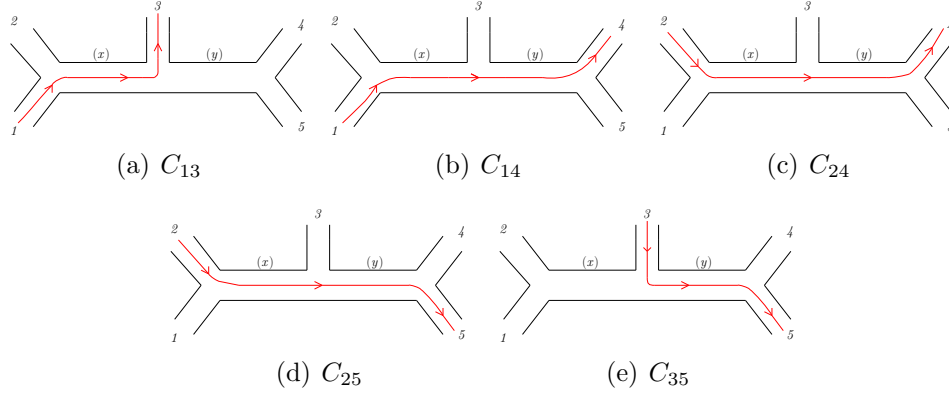


Figure 4.9: Five curves of five-leg tree fatgraph.

Again, these curves encode all the information we need to compute our amplitudes. We can also recognise the fact that all these curves change direction at least once. In this case, if a curve avoids an exit, such as C_{14} not turning left to exit the fatgraph to external edge 3, we will consider this a right turn.

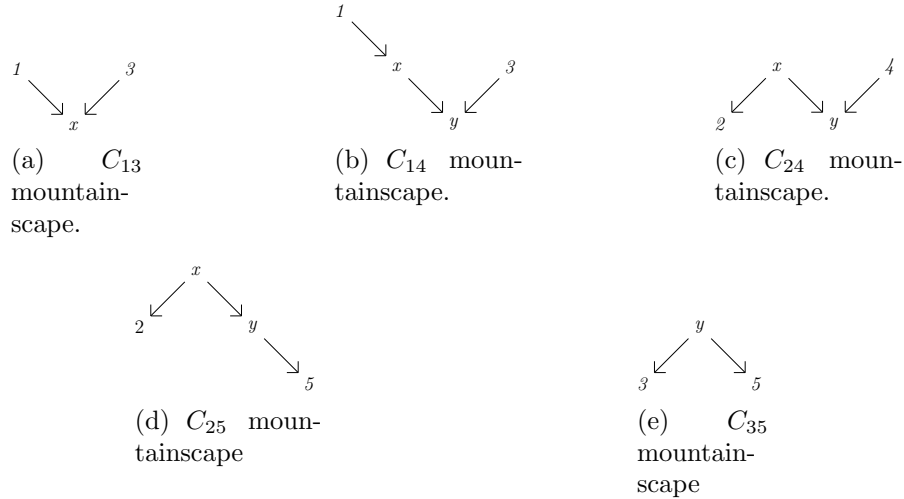


Figure 4.10: Mountainscape diagrams of five-leg fatgraph.

From figure 4.9, we can draw our mountainscape diagrams in figure 4.10, following the same rules as in the previous example. From our mountainscape diagrams we can find our asso-

ciated $\mathbf{g}$ -vectors. Now these vectors will be two dimensional, as we have two internal edges which can correspond to peaks or valleys. Our $\mathbf{g}$ -vectors are found to be,

$$\begin{aligned}\mathbf{g}_{C_{13}} &= \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \mathbf{g}_{C_{14}} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \mathbf{g}_{C_{24}} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \\ \mathbf{g}_{C_{25}} &= \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{g}_{C_{35}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix},\end{aligned}\tag{4.35}$$

which plot out the $\mathbf{g}$ -vector space:

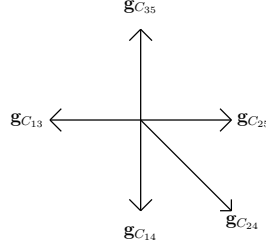


Figure 4.11: $\mathbf{g}$ -vector fan for five-leg fatgraph.

Whilst following the same sub-mountainscape rules used in figure 4.4, we achieve the following F -polynomials,

$$\begin{aligned}F_{C_{13}} &= 1 + x + ax + bx + abx, \\ F_{C_{14}} &= 1 + y + xy + bxy + axy + abxy, \\ F_{C_{24}} &= 1 + a + y + ay + axy + by + abxy + aby, \\ F_{C_{25}} &= 1 + a + b + ab + yb + yab + abxy, \\ F_{C_{35}} &= 1 + a + b + ab + yab.\end{aligned}\tag{4.36}$$

Next we will again use our F -polynomials to fill in the corresponding curve-matrices of the form,

$$M_C = \begin{pmatrix} F_{-+} & F_{+-} \\ F_{--} & F_{++} \end{pmatrix}.\tag{4.37}$$

In doing so our curve-matrices are,

$$\begin{aligned}M_{C_{13}} &= \begin{pmatrix} 1+x & x \\ x & x \end{pmatrix}, & M_{C_{14}} &= \begin{pmatrix} 1+y(1+x) & xy \\ y(x+1) & xy \end{pmatrix}, \\ M_{C_{24}} &= \begin{pmatrix} 1+y & 1+y+xy \\ y & y(1+x) \end{pmatrix}, & M_{C_{25}} &= \begin{pmatrix} 1 & 1+y \\ 1 & 1+y+xy \end{pmatrix}, \\ M_{C_{35}} &= \begin{pmatrix} 1 & 1 \\ 1 & 1+y \end{pmatrix}.\end{aligned}\tag{4.38}$$

By following the u -variable definition of,

$$u_C = \frac{F_{-+}F_{+-}}{F_{--}F_{++}} = \frac{(M_C)_{12}(M_C)_{21}}{(M_C)_{11}(M_C)_{22}},\tag{4.39}$$

we can define the u -variables of the five-leg fatgraph curves as,

$$\begin{aligned}
u_{C_{13}} &= \frac{x}{1+x}, \\
u_{C_{14}} &= \frac{y(x+1)}{1+y+xy}, \\
u_{C_{24}} &= \frac{(1+y+xy)y}{(1+y)(x+1)y} = \frac{(1+y+xy)}{(1+y)(x+1)}, \\
u_{C_{25}} &= \frac{1+y}{1+x+y}, \\
u_{C_{35}} &= \frac{1}{1+y}.
\end{aligned} \tag{4.40}$$

In order to acquire our headlight functions we must tropicalise our u -variables, which result headlight functions of the following form:

$$\begin{aligned}
\alpha_{C_{13}} &= \max(0, x) - x, \\
\alpha_{C_{14}} &= -y - \max(0, x) + \max(0, y, x+y), \\
\alpha_{C_{24}} &= -\max(0, y, x+y) + \max(0, y) + \max(0, x), \\
\alpha_{C_{25}} &= -\max(0, y) + \max(0, y, x+y), \\
\alpha_{C_{35}} &= \max(0, y).
\end{aligned} \tag{4.41}$$

Now we can perform our integration as in the previous example,

$$\mathcal{A}_{n=5} = \int dx dy (-\log Z), \tag{4.42}$$

where $-\log Z = -X_{13}\alpha_{C_{13}} - X_{14}\alpha_{C_{14}} - X_{24}\alpha_{C_{24}} - X_{25}\alpha_{C_{25}} - X_{35}\alpha_{C_{35}}$.

What we will do is split this integral up into the five regions cut of by the five cones, and integrate each pair of kinematic variables, along with their corresponding headlight functions, over their corresponding cones. The headlight functions are linear functions within their own cones.

$$\begin{aligned}
\mathcal{A}_{n=5} &= \int dx dy (-\log Z) + \\
&\int_0^\infty \int_0^\infty dx dy (-\log Z) + \int_0^\infty \int_{-\infty}^0 dx dy (-\log Z) + \\
&\int_{-\infty}^0 \int_{-\infty}^0 dx dy (-\log Z) + \int_{-\infty}^0 \int_0^{-y} dx dy (-\log Z) + \\
&\int_0^{-\infty} \int_{-y}^\infty dx dy (-\log Z).
\end{aligned} \tag{4.43}$$

Over each set of bounds, the only non zero headlight functions are the ones with the associated $\mathbf{g}$ -vectors at the boundaries. An example of this will now be shown for one of the more non

trivial cases, the cone bordered by the C_{25} and C_{24} $\mathbf{g}$ -vectors.

$$\int_{-\infty}^0 \int_0^{-y} dx dy (-\log Z) = \int_{-\infty}^0 \int_0^{-y} \exp(-X_{24}\alpha_{C_{24}} - X_{25}\alpha_{C_{25}}) dx dy. \quad (4.44)$$

This is as every other headlight function equates to zero between these bounds. We can see an example of this for $\alpha_{C_{24}} = -\max(0, y, x+y) + \max(0, y) + \max(0, x)$, where after applying the bounds we are left with $\alpha_{C_{24}} = -(-y+y) + 0 + x = x$, which shows that the headlight function survives. For a vanishing headlight function we will use $\alpha_{C_{14}} = -y - \max(0, x) + \max(0, y, x+y)$, which after applying the bounds leaves us with $\alpha_{C_{14}} = -y - x + (x+y) = 0$. After applying these bounds we simply integrate with the remaining headlight functions as we did for the $n = 4$ point amplitude and we are finally left with our amplitude,

$$\mathcal{A}_{n=5} = \int dx dy (-\log Z) = \left(\frac{1}{X_{13}X_{1,4}} + \frac{1}{X_{35}X_{13}} + \frac{1}{X_{14}X_{24}} + \frac{1}{X_{25}X_{35}} + \frac{1}{X_{24}X_{25}} \right). \quad (4.45)$$

Chapter 5

Connection Between Associahedron and Fatgraph

We should note that in order to connect these two methods, we need to revert the fatgraph method back to the massless case, to allow for the same factorisation properties in the associahedron method.

5.0.1 The Fatgraph Feynman-Fan, as the Kinematic Associahedron Dual

Now the last thing we need to do is to show how our fatgraph method is connected to our associahedron, using the dual method as shown in [1]. In section 2.7.3, we showed every convex polytope has a corresponding dual polytope, i.e. $\mathcal{A}$ has an $\mathcal{A}^*$. As we can see in section 7.1,

$$\underline{\Omega}(\mathcal{A}) = \text{Vol}(\mathcal{A}^*) \quad (5.1)$$

For us this now means that the canonical form of the associahedron is the volume of the dual associahedron. As the canonical form is determined by the amplitude m_n , this suggests that the amplitude is the volume of the dual,

$$m_n = \text{Vol}(\mathcal{A}_n^*). \quad (5.2)$$

As in section 2.7.3 we showed how the dual polytopes are defined in projective space, we now embed our subspace H_n in projective $\mathbb{P}^{n-3}(\mathbb{R})$, along with a chosen basis of Mandelstam variables, $X_{i'_1, j'_1}, \dots, X_{i'_n, j'_n}$, which gives us our subspace,

$$\mathbb{Y} = (1, X_{i'_1, j'_1}, \dots, X_{i'_n, j'_n}) \in \mathbb{P}^{n-3}(\mathbb{R}). \quad (5.3)$$

The 1 corresponds to a zeroth coordinate, as the coordinates are projective.

We can also obtain the associahedron facets in projective coordinates. The facets of the associahedron are when $X_{i,j} = 0$, so we rewrite this in the form $W_{i,j} \cdot Y = 0$, where $W_{i,j}$ is a dual vector.

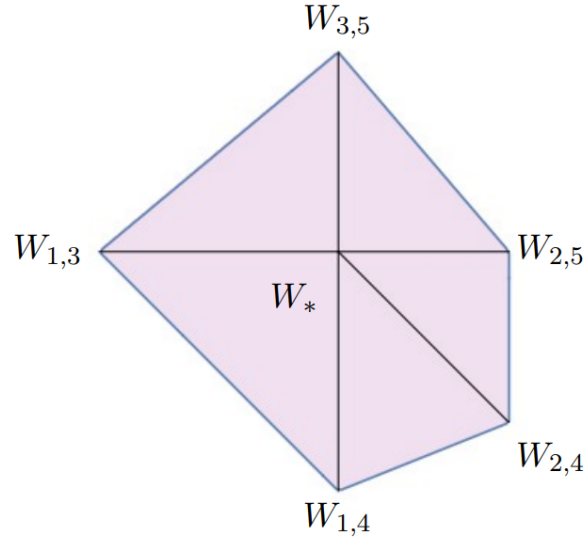


Figure 5.1: One triangulation of the $\mathcal{A}_{n=5}^*$ dual associahedron [1].

We can show an example of this for our $n = 5$ case, where the basis is $Y = (1, X_{1,3}, X_{4,5})$. We can write down the following equalities,

$$Y \cdot W_{2,5} = X_{2,5} = c_{13} + c_{14} - X_{1,3} = (c_{13} + c_{14}, -1, 0) \cdot W. \quad (5.4)$$

From this we can simply read off $W = (c_{13} + c_{14}, -1, 0)$.

The following coordinates are in reference to our typical $n = 5$ associahedron, in figure 3.4,

$$\begin{aligned} W_{1,3} &= (0, 1, 0) \\ W_{3,5} &= (c_{14} + c_{24}, 0, -1) \\ W_{2,5} &= (c_{13} + c_{24}, -1, 0) \\ W_{2,4} &= (c_{13}, -1, 1) \\ W_{1,4} &= (0, 0, 1). \end{aligned} \quad (5.5)$$

The coordinates of our dual vectors, are the vertex points of the dual associahedron, $\mathcal{A}_n^*$, as seen in figure 5.1. We can also see that our projective coordinates are just projective version of our $n = 5$ $\mathbf{g}$ -vectors from equation 4.35. This means that when we integrate over our Feynman-fan to solve for the form of our amplitude, we are calculating the volume of our dual associahedron! The fatgraph method contains all the complexities of the Kinematic associahedron, such as poles at boundaries, showing locality, and factorisation, showing unitarity, when taken in the massless case.

5.1 The Fatgraph Restores All Feynman Diagram Triangulation of the Associahedron

This section explains the triangulation equivalence of one Feynman diagram, to one fatgraph, and how that one fatgraph can give back the amplitude of all the triangulations.

We will calculate, by triangulation and summation over the volume of each piece, the volume of $\mathcal{A}^*$. We can notice that our dual associahedron, as seen in figure 5.1, is a simple polytope, as it is two-dimensional, and each vertex is adjacent to two edges. This means that each facet is a simplex. This is the same as saying that $\mathcal{A}_n$ is a simple polytope, and in this case the dual is triangulated as follows:

- We will place a reference point, which we will label W_* , in the interior of our dual polytope.
- A convex hull is taken for each facet of the dual, resulting in a simplex, corresponding to each facet.
- The union of all of these simplices forms a triangulation of the dual.

We will now formalise these steps. A facet of the dual $\mathcal{A}^*$ will be denoted by Z . Z is adjacent to vertices labelled $W_{i_{n-3},j_{n-3}}$, which correspond to the propagators $X_{i_1,j_1}, \dots, X_{i_{n-3},j_{n-3}}$, respectively. We then take the convex hull of Z , with the point W_* , and after taking the union of convex hulls over all facets, we are left with a triangulation of the dual associahedron, whose volume is, as we know, the sum over each simplex volume.

We can now write the equation,

$$\text{Vol}(\mathcal{A}_n^*) = \sum_{\text{vertex } Z} \text{Vol}(W_*, W_{i_1,j_1}, \dots, W_{i_{n-3},j_{n-3}}), \quad (5.6)$$

which we now relate to equation 3.27,

$$\begin{aligned} \text{Vol}(\mathcal{A}_n^*) &= \sum_{\text{vertex } Z} \text{Vol}(W_*, W_{i_1,j_1}, \dots, W_{i_{n-3},j_{n-3}}), \\ &= \sum_{\text{vertex } Z} \frac{\text{sign}(Z) \langle W_* W_{i_1,j_1} \dots W_{i_{n-3},j_{n-3}} \rangle}{(Y \cdot W_*) \prod_{a=1}^{n-3} (Y \cdot W_{i_a,j_a})}. \end{aligned} \quad (5.7)$$

In the above equation, $\text{sign}(Z)$ is dependent on the orientation of the adjacent vertices, $W_{i_1,j_1} \dots W_{i_{n-3},j_{n-3}}$, relative to the inherited orientation. The anti-symmetry in the numerator, from the $\text{sign}(Z)$, is cancelled out but the anti-symmetry of the determinant, $\langle \dots \rangle$, in the denominator. We also note that the sum is not dependent of the our chosen point W_* . The $\text{sign}(Z)$ term in the above equation correlates the the $\text{sign}(Z)$ term in equation 3.27, where in equation 3.27, Z is the corresponding vertex in $\mathcal{A}_n$.

What is left is to show that for a specific reference point W_* , the expected amplitude m_n , is equivalent to equation 5.7. As we know m_n we can choose an appropriate point, $W_* = (1, 0, \dots, 0)$, which means the numerator in equation 5.7 is equivalent for all choices of Z . As $Y \cdot W_{i_a,j_a} = X_{i_a,j_a}$, we can formulate the following,

$$\begin{aligned} \langle W_* W_{i_1,j_1} \dots W_{i_{n-3},j_{n-3}} \rangle &= \frac{\partial(X_{i_1,j_1}, \dots, X_{i_{n-3},j_{n-3}})}{\partial(X'_{i'_1,j'_1}, \dots, X'_{i'_{n-3},j'_{n-3}})} \\ &= \frac{\text{sign}(Z)}{\text{sign}(Z')}. \end{aligned} \quad (5.8)$$

The last equality comes from equation 3.29, and the primed labels are from equation 5.3. Now this means the numerators for all Z 's of equation 5.7 are equal to $\text{sign}(Z')$, which we

let equal to one. Now, the last thing we need to do is substitute $(Y \cdot W_*) = 1$, as well as $(Y \cdot W_{i,a,j_a} = X_{i_a,j_a})$ into equation 5.7, to obtain,

$$\text{Vol}(\mathcal{A}_n^*) = \sum_{\text{planar } g} \frac{1}{\prod_{a=1}^{n-3} X_{i_a,j_a}}, \quad (5.9)$$

where we have also replaced Z , with g . The last thing to say is,

$$\text{Vol}(\mathcal{A}_n^*) = m_n, \quad (5.10)$$

as desired.

Recall partial triangulations from section 3.2.1. Each convex hull corresponding to each fact of $\mathcal{A}^*$ is a partial triangulation, and we can see that the volume of each partial triangulation corresponds, uniquely to a term of the total sum in equation 5.9. As we saw in chapter 1, each labelling of a Feynman diagram corresponds to an individual term of equation 5.9. As well as this, each partial triangulation amplitude corresponds to a unique amplitude from a specific cone of the Feynman-fan, as we saw in chapter 4. All of this means that for one specific labelling of a Feynman diagram, i.e. one partial triangulation of our dual associahedron $\mathcal{A}^*$, we can “fatten” this labelling and obtain a Feynman-fan, from curves through our fatgraph, which is equivalent to the volume of $\mathcal{A}_n^*$. In conclusion we can say, in the original equivalence formula, when all in massless $\text{Tr}(\phi)^3$ theory,

$$\boxed{\underline{\Omega(\mathcal{A})} = \text{Vol}(\mathcal{A}^*) = \int_{\text{cone}} d^E t e^{-S} = m_n.} \quad (5.11)$$

Chapter 6

Conclusions

Throughout this work, there has been an exploration into a new geometric method of obtaining scattering amplitudes, in particular for $n = 4$ and $n = 5$ particle scattering. The reason for this new method is to try to reduce complications in the current Feynman diagram method. The new method requires scalar, planar amplitudes, with a specific colour ordering. Original work was done by condensing material from a series of papers, in particular [3], [1], and [2], and showing explicit connections between each.

The Associahedron method was shown to encode the tree-level Feynman amplitude into the differential $n - 3$ -forms, for n particle scattering. This was done by constructing a kinematic space, with axes of kinematic variables, which are generalised Mandelstam variables. The positive region of this space, along with the intersection of the subspace $\mathcal{H}_n$, where $\dim \mathcal{H}_n = n - 3$, is the kinematic associahedron. The n -form in kinematic space, when pulled back to $\mathcal{H}_n$, encodes within it the amplitude.

The Fatgraph method was shown to only require one Feynman diagram labelling, “triangulation”, which can be then used to regain all tree-level Feynman diagrams, from the boundary curves encoded in the surface, with an original example shown for $n = 4$ particle scattering.

We finally demonstrated how the Kinematic Associahedron and Fatgraph are dual to each other, where in massless $\text{Tr}(\phi^3)$ theory, the fatgraph method contains all the complexities of the Kinematic associahedron, such as poles at boundaries, showing locality, and factorisation, showing unitarity. We also showed our original equivalence formula, directly connecting the calculations of the amplitude from the volume of the dual associahedron, to the curve integral, in equation 5.11.

There is much room for continuation of this project, as what is included here is only scratching the surface of the complete geometric picture of scattering amplitudes. Possibilities can include; looking into the amplituhedron, the forefather of the associahedron, which actually is the associahedron for $N = 4$ SYM, as shown in [1]. Another possibility is looking more into [7], where zeros not found in Feynman diagram amplitudes are explored, as well as also looking into the connection between the associahedron and Newton polytopes.

Chapter 7

Appendix

7.1 Mathematic Background

This appendix subsection provides useful definitions which are used in section 2.

7.1.1 Complete Mathematical Definition of Algebraic Varieties and Algebraic sets

This section follows the definitions layed out in Robert Hartshorne's "Algebraic Geometry" [9].

Definition - Complex Projective Space: Complex projective space, denoted $\mathbb{P}$, following notation in [3], is defined as the set of all 1-dimensional complex-linear subspaces of $\mathbb{C}^{n+1}$, with the projection mapping of,

$$\pi : \mathbb{C}^{N+1} \setminus 0 \rightarrow \mathbb{P}^N. \quad (7.1)$$

Definition: Algebraically Closed Fields: An algebraically closed field is a field k , in which every polynomial in k has a root in k .

The following is the setup for our definition of a projective algebraic variety

Let k be a fixed algebraically closed field. *Projective n -space* over k is defined to be the set of equivalence classes of $(n + 1)$ tuples $(a_0, \dots, a_n)$ of elements of k , not all zero, under the equivalence relation of

$$(a_0, \dots, a_n) \sim \lambda(a_0, \dots, a_n), \quad (7.2)$$

for all $\lambda \in k$, such that $\lambda \neq 0$. If p is an element of $\mathbb{P}^n$, it is called a point. For a point P , any $(n+1)$ tuple $(a_0, \dots, a_n)$ in the equivalence class of p is called a set of homogeneous coordinates for p . We will let S be the polynomial ring $k[x_0, \dots, x_n]$. For $f \in S$, where f is a polynomial, it cannot be used to define a function on $\mathbb{P}^n$, as homogeneous coordinates are not unique. In the case that f is a homogeneous polynomial of degree d , $f(\lambda a_0, \dots, \lambda a_n) = \lambda^d f(a_0, \dots, a_n)$ which means the property of f being or not being zero only depends on the equivalence class on $(a_0, \dots, a_n)$. Now as the equivalence class $(a_0, \dots, a_n)$ is the only determining factor of whether f is zero or not, meaning that f gives a function from $\mathbb{P}^n$ to $\{0, 1\}$, for the function $f(p) = 0$, if $f(a_0, \dots, a_n) = 0$. On the other hand, f gives a function from $\mathbb{P}^n$ to $\{0, 1\}$ for

the function $f(p) = 1$, if $f(a_0, \dots, a_n) \neq 0$. What all of this means that we can talk about the *zeros* of a polynomial which is homogeneous. In generic term we can call the zeros of a homogeneous polynomial f , $Z(f)$, where

$$Z(f) = \{p \in \mathbb{P}^n | f(p) = 0\}. \quad (7.3)$$

For T being the set of homogeneous elements in S , the *zero set* of T is defined to be

$$Z(T) = \{p \in \mathbb{P}^n | f(p) = 0 \ \forall f \in T\}. \quad (7.4)$$

Definition - Algebraic Set: An Algebraic Set is a subset $Y \subset \mathbb{P}^n$ where there exists of homogeneous elements T , such that $Y = Z(T)$.

Definition: Projective Algebraic Variety: A projective algebraic variety, is an irreducible algebraic set in $\mathbb{P}^n$.

7.1.2 Definition - Residue Operator

Let ω be a meromorphic differential form on X , where C is an irreducible sub variety of X and z is a holomorphic coordinate whose zero set, $z = 0$, parametrises C . All other holomorphic coordinates will be denoted collectively as u . This implies that a *simple pole* of ω at C is a singularity of the form:

$$\omega(u, z) = \omega'(u) \wedge \frac{dz}{z} + \dots, \quad (7.5)$$

where terms smooth in the small z limit are disregarded, and where $\omega'(u)$ is a non-zero meromorphic form defined locally on the boundary component. The residue operator of ω at C is defined,

$$Res_C \omega := \omega'(locally). \quad (7.6)$$

If there does not exist such a simple pole the residue operator is defined to be zero.

7.1.3 Forms on Projective Spaces

We will let η be a rational m -form on $\mathbb{C}^{m+1}$, which is viewed as follows,

$$\eta = \sum_{I=0}^m f_I(Y) dY^0 \wedge \dots \wedge \widehat{dY^I} \wedge \dots \wedge dY^m, \quad (7.7)$$

where in the regular form notation, $\widehat{dY^I}$ means we omit dY^I . $f_I(Y)$ is a rational function. This means η is the pull back of a different rational m -form, which is defined on $\mathbb{P}^m$, if and only if both of the following are satisfied,

1. η is homogeneous of degree 0, meaning that $f_I(Y)$ is homogeneous of degree $-m$.
2. $\langle \eta, \varepsilon \rangle = 0$, where ε is the Euler vector field, which is a radial vector field, $\varepsilon = \sum_{I=0}^m Y^I \partial_I$.

We can solve (2) and be left with, as given in [3],

$$\eta = m! \cdot g(Y) \sum_{I=0}^m (-1)^I Y dY^0 \wedge \dots \wedge \widehat{dY^I} \wedge \dots \wedge dY^m, \quad (7.8)$$

for a rational functional $g(Y)$, of degree $-m - 1$. We can formulate,

$$\sum_{I=0}^m (-1)^I Y dY^0 \wedge \cdots \wedge \widehat{dY^I} \wedge \cdots \wedge dY^m = \frac{1}{m!} \langle Y d^m Y \rangle, \quad (7.9)$$

which means,

$$\langle Y d^m Y \rangle = m! \sum_{I=0}^m (-1)^I Y dY^0 \wedge \cdots \wedge \widehat{dY^I} \wedge \cdots \wedge dY^m. \quad (7.10)$$

Now we can write an arbitrary form on $\mathbb{P}^m$ as,

$$\eta = g(Y)\omega, \quad (7.11)$$

where $\omega := \langle Y d^m Y \rangle$. We can call ω a *standard measure*, and if η is the canonical form of a pseudo-positive geometry $(\mathbb{P}^m, \mathcal{A})$, $g(Y)$ is called the *canonical rational function* of $(\mathbb{P}^m, \mathcal{A})$, which is written in the literature as $\underline{\Omega}(\mathcal{A})$, and can also be defined as $\underline{\Omega}(\mathcal{A}) = \Omega(\mathcal{A}) / \langle Y d^m Y \rangle$.

An important note to make is that an equivalent, alternative description of a form η on projective space, is that the form η must be invariant under *local* $GL(1)$ transformations.

7.1.4 Note on Positive Geometries

For two positive geometries $\mathcal{A}$ and $\mathcal{B}$, we can note that

$$\Omega(\mathcal{A} \times \mathcal{B}) = \Omega(\mathcal{A}) \times \Omega(\mathcal{B}), \quad (7.12)$$

where $\Omega(\mathcal{A}) \times \Omega(\mathcal{B})$ is the wedge product of the two canonical forms.

7.2 Kinematic Associahedron

7.2.1 Vertex Coordinates For the Kinematic Associahedron

We want to connect our vertex labelling to our kinematic variable labelling of our associahedron. We will first consider of a mutation of $Z \rightarrow Z'$, which can be seen in the figure 3.1 We can mutate $X_{i,k}$ to $X_{j,l}$, which from equation 3.24, we obtain

$$X_{j,l} = X_{j,k} + X_{i,l} - X_{i,k} + \sum_{\substack{i \leq a < b \\ k \leq b < l}} c_{ab}. \quad (7.13)$$

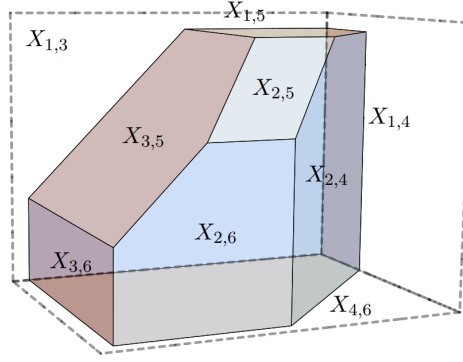
Which computes $X_{j,l}$ from the basis of Z .

The vertex coordinates of the $n = 5$ particle scattering kinematic associahedron $\mathcal{A}_{n=5}$, from figure

with the basis $Y = (1, X_{13}, X_{14})$,

$$\begin{aligned} Z_A &= (1, 0, 0) \\ Z_B &= (1, 0, c_{14} + c_{24}) \\ Z_C &= (1, c_{13} + c_{14}, c_{14} + c_{24}) \\ Z_D &= (1, c_{13} + c_{24}, c_{14}) \\ Z_E &= (1, c_{13}, 0). \end{aligned} \quad (7.14)$$

7.2.2 $n = 6$ Kinematic associahedron

Figure 7.1: $n = 6$ associahedron

7.2.3 U -Variables

U -variables satisfy the non-linear equation

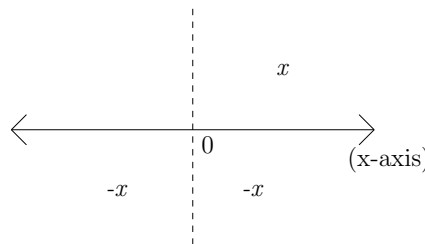
$$u_C + \prod_D u_D^{n(C,D)} = 1, \quad (7.15)$$

where $n(C, D)$ is the intersection number between curves C and D . The intersection number is defined the minimum number of times the two curves can intersect, while keeping their homotopy classes unchanged (i.e. how many times they intersect while deforming them to intersect as little as possible, while keeping the same endpoints[10]). This equation is not necessary for our calculations, but is the background for its use, and is important for possible extensions of this project.

7.3 Fatgraph Method

7.3.1 Another Way to View the Headlight Function

The origin of these headlight function are rooted in tropical geometry and the amplitudehedron, most of which is outside the scope of this project. However, from this we can view our headlight functions as follows, with specific example from 4.26.

Figure 7.2: Visualisation of $\alpha_{C_{13}} = \max(0, x) - x$.

What we can see here is when $x \geq 0$, the headlight function gives us two values, an x term and a $-x$ term (the above figure is just for visualisation purposes, as all x 's in reality lie directly along x-axis, which is our 1-dimensional curve space in this instance). This means in the section of curve space where $x \geq 0$, we have no terms as the x and $-x$ will cancel out. As there is only a $-x$ term, our headlight function in curve space is seen in figure 7.3. When we perform our integration, our $\alpha_{C_{13}}$, which is coupled to our X_{13} kinematic variable, is then only integrated over x between $(-\infty, 0]$.

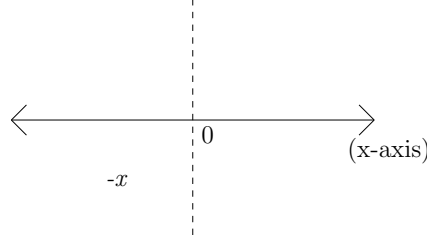


Figure 7.3: Final visualisation of $\alpha_{C_{13}} = \max(0, x) - x$.

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